

Stopped Experiment Complexity for Stochastic Bandit Convex Optimization

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Abstract

We introduce a stopped cost–progress ratio, a multistep analogue of the information ratio, for stationary stochastic bandit convex optimization. At posterior-risk scale b , a variable-length observation-adaptive experiment incurs cost $b^2 \mathbb{E}N + b \mathbb{E}R$. Its progress is mutual information about one near-optimal action retained across experiments at that scale, plus credit for the first constant-factor reduction in posterior Bayes risk. The stopped chain rule expresses transcript information as the sum of the conditional information gains collected along the experiment, while query selection, continuation, and stopping are optimized jointly. For a scale-calibrated non-hitting one-query experiment, the resulting ratio is comparable, up to constants, to the target-specific information-ratio form ℓ^2/i .

The measurable-policy profile $\Lambda^{\text{policy}}(d, T)$, the infimum of the worst-state ratios achievable by measurable state-dependent experiment rules over the active scale window, yields $\tilde{O}(\sqrt{d\{1 + \Lambda^{\text{policy}}(d, T)\}T})$ regret. Conversely, for every $0 < \lambda < \Lambda^{\text{state}}(d, B)$, some active initial prior at a scale $b \in [B^{-1}, 1]$ forces $\Omega(\sqrt{d\lambda n})$ regret at $n = \lfloor d\lambda/(4b^2) \rfloor$, whenever $d\lambda/b^2 \geq 8$. At one state, negative and nonnegative offsets give the two local Bellman certificates. A cost-sensitive Bellman–Fano factorization derives the latter from the cost per unit full-model information and the entropy required for a posterior-risk contraction. Gaussian measurable implementation identifies the statewise and policy profiles; denote their common value by $\Lambda(d, B)$. For nondegenerate class sequences ($\liminf_{d \rightarrow \infty} \Lambda(d, 2) > 0$), Gaussian Bayes–minimax duality gives

$$\beta_{\mathfrak{F}} = \frac{1 + \alpha_{\text{stop}}}{2} = \frac{1 + \alpha_{\text{hit}}}{2},$$

where $\beta_{\mathfrak{F}}$ is the polynomial dimension exponent of minimax regret normalized by $\sqrt{T}$, and α_{stop} and α_{hit} are the corresponding information-assisted and hit-only profile exponents. Thus first-hit continuation suffices at the exponent level, while mutual information supplies chain-rule-additive credit for informative non-hitting blocks; equality of the two finite profiles is not asserted.

For the dimple family of Bakhtiari, Lattimore, and Szepesvári, both one-query and fresh-uniform-direction restricted stopped values are $\Omega(d^2)$, whereas observation-adaptive continuation attains $\tilde{O}(d)$ for both profiles. At $\varepsilon = d^{-1}$ this implies the class-level minimax upper bound $\tilde{O}(\min\{T/d, d\sqrt{T}\})$. Determining the stopped profile for the full convex class remains equivalent, at the exponent level, to closing the minimax dimension gap.

1 Introduction

Let $C \subset \mathbb{R}^d$ be a convex body. A continuous convex loss $F : C \rightarrow [0, 1]$ is drawn once and remains fixed. At round t , a learner chooses $X_t \in C$ and observes

$$O_t = F(X_t) + \xi_t, \quad \xi_t \stackrel{\text{iid}}{\sim} N(0, 1).$$

Its cumulative regret is

$$R_T = \sum_{t=1}^T \{F(X_t) - \min_{x \in C} F(x)\}.$$

This is stationary stochastic bandit convex optimization (BCO), equivalently noisy zeroth-order convex optimization with cumulative rather than terminal loss. In the nontrivial long-horizon regime, the best general upper bound is $\tilde{O}(d^{3/2}\sqrt{T})$ (Lattimore and György, 2023; Fokkema et al., 2024; Carpentier, 2025). Rajaraman’s Corollary 1.2 gives $\Omega(d^{5/4}\sqrt{T/\log(ed)})$ regret on the Euclidean ball when $T \gtrsim d^3/\log(ed)$ (Rajaraman, 2026). This is the first general lower bound with dimension dependence strictly larger than the $\Omega(d\sqrt{T})$ inherited from linear bandits (Dani et al., 2008; Shamir, 2013). The numerical gap is therefore narrow but conceptually unresolved: what feature of a convex function costs more than learning an unknown linear direction? Neither endpoint is known to be sharp. Throughout, the *full bounded convex class* consists of all continuous $[0, 1]$ -valued convex losses on C .

From one-query ratios to stopped experiments. The relevant distinction is the unit over which the local tradeoff is optimized. The information ratio compares squared conditional regret with the information obtained from one randomized query (Russo and Van Roy, 2016, 2018; Lattimore and György, 2021b), while the decision–estimation coefficient (DEC) balances one decision’s loss against the statistical separation generated by its observation (Foster et al., 2021, 2022, 2023). The algorithmic information ratio (AIR) supplies an exact posterior-coordinate identity whose information potential telescopes across rounds (Xu and Zeevi, 2025). The associated algorithms are fully sequential and may re-optimize after every observation; the point is that their canonical local coefficient is evaluated on one native query–observation pair.

In BCO, an observation can reveal the geometry in which the next probe should be placed. A large one-query coefficient may therefore measure the difficulty of choosing a direction before seeing the data, rather than the difficulty of an experiment whose later directions depend on earlier observations. This issue is concrete for the smooth dimple family of Bakhtiari et al. (2025), whose one-query coefficient is of order d^2 .

We therefore optimize the tradeoff over a variable-length adaptive experiment. Let $r_B(h)$ be posterior Bayes risk after history h , fix a scale b with $r_B(h) \asymp b$, and draw the canonical near-optimal target A_b° once for all later experiments at that scale. Let Z be a complete experiment stopped no later than the first history at which $r_B \leq \gamma b$, let N_Z be its length and R_Z its exploratory regret, and let M denote this first hit. If ℓ_t is the conditional expected regret of query t and i_t its conditional information gain about A_b° , then the tower property and stopped chain rule give

$$b^2\mathbb{E}[N_Z \mid h] + b\mathbb{E}[R_Z \mid h] = \mathbb{E}\left[\sum_{t \leq N_Z} (b^2 + b\ell_t) \mid h\right], \quad I(A_b^\circ; Z \mid h) = \mathbb{E}\left[\sum_{t \leq N_Z} i_t \mid h\right]. \quad (1)$$

The information-assisted stopped value is therefore

$$\Pi_b(h; A_b^\circ) = \inf_Z \frac{\mathbb{E}[\sum_{t \leq N_Z} (b^2 + b\ell_t) \mid h]}{\mathbb{E}[\sum_{t \leq N_Z} i_t \mid h] + d\mathbb{P}(M \mid h)}. \quad (2)$$

It is a ratio of accumulated cost to accumulated progress, not a sum of separately optimized one-step ratios. For one non-hitting query with expected regret $\ell \asymp b$, the corresponding one-query ratio satisfies $(b^2 + b\ell)/i \asymp \ell^2/i$, recovering the target-specific information-ratio scale. More generally, Π_b jointly optimizes an observation-adaptive query, continuation, and stopping policy before comparing

its accumulated expected cost and progress. For a candidate ratio level $q > 0$ and a finite query cap L , the ratio–offset form is

$$\inf_{Z: N_Z \leq L} \mathbb{E} \left[\sum_{t \leq N_Z} \{b^2 + b\ell_t - qi_t\} - qd\mathbf{1}_M \mid h \right], \quad (3)$$

whose belief-state value obeys a controlled Bellman optimal-stopping recursion. The unrestricted value is the uncapped limit of these Bellman problems.

To isolate the terminal progress mechanism, we also delete the information increments and study the hit-only ablation

$$\Pi_b^{\text{hit}}(h) = \inf_Z \frac{b^2 \mathbb{E}[N_Z \mid h] + b \mathbb{E}[R_Z \mid h]}{d\mathbb{P}(\mathbf{M} \mid h)}. \quad (4)$$

The hit records a completed risk contraction, whereas target information accumulates partial progress across non-hitting blocks. We prove that the two optimized profiles have the same polynomial exponent. Equality of their finite optimized values is neither required nor proved, and their accounting differs strictly on fixed non-hitting blocks. In this precise sense, adaptive continuation is the rate-bearing primitive and retained information provides a dense compositional currency.

The two-sided principle. For an active posterior state $s = (Q, b)$ and a first-hit experiment $\mathcal{E}$, write $\rho(s, \mathcal{E})$ for the ratio in (2). Its pointwise stopped value is $\Pi(s) := \inf_{\mathcal{E}} \rho(s, \mathcal{E}) = \Pi_b(Q; A_b^\circ)$. Over active scales $B^{-1} \leq b \leq 1$, the statewise profile $\Lambda^{\text{state}}(d, B)$ is the worst pointwise value $\sup_s \Pi(s)$, whereas $\Lambda^{\text{policy}}(d, B)$ is the infimum of the uniform bounds achievable by universally measurable state-dependent experiment selectors. Here and below this introductory notation suppresses the outer worst case over the known body. Thus $\Lambda^{\text{state}} \leq \Lambda^{\text{policy}}$.

The mathematical core is the pair of finite-value reductions

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \tilde{O} \left(\sqrt{d\{1 + \Lambda^{\text{policy}}(d, T)\}T} \right), \quad (5)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq c_\gamma \sqrt{d\lambda n}, \quad n = \left\lfloor \frac{d\lambda}{4b^2} \right\rfloor. \quad (6)$$

Here λ is not another profile. For every $0 < \lambda < \Lambda^{\text{state}}(d, B)$, some active initial state at a scale $b \in [B^{-1}, 1]$ has $\Pi(s) > \lambda$, and (6) holds whenever $d\lambda/b^2 \geq 8$. Thus Λ^{policy} is a uniform upper certificate, while λ is a test level strictly below the stopped value of one hard state; neither $\lambda = \Lambda$ nor attainment of the profile supremum is asserted.

The upper reduction glues variable-length experiments rather than merely concatenating them: it retains one target through repeated visits to a risk scale, telescopes the information accumulated across stopped blocks, and controls returns after a successful contraction. The converse reverses this construction. Convex averaging turns a low-regret policy into a terminal recommendation, and stopping at the first risk reduction makes it an admissible experiment. This first-hit reduction makes the stopped profile necessary, rather than only a sufficient upper-bound certificate.

The two directions initially have different quantifier orders (with the scale window and outer supremum over known bodies suppressed):

$$\underbrace{\sup_s \inf_{\mathcal{E}} \rho(s, \mathcal{E})}_{\text{statewise profile } \Lambda^{\text{state}}} \leq \underbrace{\inf_{\sigma \text{ measurable}} \sup_s \rho(s, \sigma(s))}_{\text{policy profile } \Lambda^{\text{policy}}}. \quad (7)$$

This is not a game-theoretic minimax gap: σ may choose a different experiment at every state. The remaining issue is whether the statewise infima can be implemented measurably without enlarging the uniform profile. For the continuous Gaussian model, we prove equality of the two profiles using universally measurable ϵ -optimal selectors, without requiring a common deterministic query cap. A direct finite-horizon Bayes–minimax argument also gives

$$\inf_{\pi} \sup_f \mathbb{E}_f R_T = \sup_Q \inf_{\pi} \mathbb{E}_Q R_T. \quad (8)$$

Under Gaussian feedback the measurable closure therefore permits the notation

$$\Lambda(d, B) := \Lambda^{\text{state}}(d, B) = \Lambda^{\text{policy}}(d, B) \quad (9)$$

for the common information-assisted profile. These two closure results turn (5)–(6) into an exact exponent identity. For every nondegenerate class sequence, meaning $\liminf_{d \rightarrow \infty} \Lambda(d, 2) > 0$, the common stopped exponent satisfies

$$\boxed{\beta_{\mathfrak{F}} = \frac{1 + \alpha_{\text{stop}}}{2} = \frac{1 + \alpha_{\text{hit}}}{2}.} \quad (10)$$

Here $\beta_{\mathfrak{F}}$ is the polynomial dimension exponent of minimax regret over polynomial horizons, and α_{hit} is defined from the hit-only profile in the same inverse-polynomial scale windows as α_{stop} . The statement is exact at the exponent level; it is not a pointwise equality between every finite-horizon regret value and either local ratio.

Contributions. First, we prove two-sided finite-value transfers between global regret and local stopped experiments. A uniformly bounded measurable-policy profile yields a regret upper bound, while a large statewise value yields a regret lower bound through a first-hit online-to-batch converse. Measurable implementation and Gaussian Bayes–minimax duality close the two quantifier gaps and give the exact exponent identity for both the information-assisted and hit-only profiles.

Second, we expose a common Bellman mechanism behind the two local directions. Ratio–offset equivalence turns the accumulated cost–progress ratio into an information-penalized optimal-stopping problem: a negative value constructs an experiment, whereas nonnegativity for every cap certifies a hard state. We further factor one such lower certificate into a Bellman bound on the cost of full-model information and a Fano entropy barrier for posterior-risk contraction. Globally, one persistent target converts the sum of blockwise information gains into a telescoping potential, while first hits and return times are amortized across geometric risk scales. The generic fixed-index and Bellman–Fano recursions come from Xu (2026a); the new content here is the BCO cost, sublevel-set target, contraction entropy, first-hit credit, risk-scale gluing, and the converse that makes the resulting local quantity necessary at the exponent level.

Third, building on the one-query obstruction of Bakhtiari et al. (2025), we show that its $\Omega(d^2)$ barrier persists for arbitrarily long fresh-direction experiments with adaptive radii, but falls to $\tilde{O}(d)$ when observations orient subsequent probes. The uniform construction also yields a new family-specific minimax upper bound.

Adaptive continuation. Our geometric result shows that the stopped value can be polynomially smaller than its one-step restriction. On the dimple family, at $\varepsilon = d^{-1}$ and $b \asymp d^{-1}$,

$$\Pi_b^{[1]} = \Omega(d^2), \quad \Pi_b^{\text{fresh}} = \Omega(d^2), \quad \Pi_b = \tilde{O}(d). \quad (11)$$

Here $\Pi_b^{[1]}$ allows one query, while Π_b^{fresh} allows adaptive radii but uses a fresh uniform direction at each query. The unrestricted experiment first learns a weak direction, amplifies it by tangent finite differences, and then contracts its angular error geometrically. The one-step obstruction therefore loses a factor $\tilde{\Omega}(d)$ once the experiment may choose later directions from earlier observations. Uniformity over all posteriors and active scales also gives a class-level $\tilde{O}(d\sqrt{T})$ minimax upper bound for this dimple family. This is not an upper bound for the full convex class, and no matching lower bound for the dimple family is claimed.

What remains open. The dimple theorem shows that a large one-query coefficient need not describe the global problem. The central remaining question is therefore to determine the unrestricted stopped profile of the full convex class. Current regret bounds place its exponent between $3/2$ and 2 , with neither endpoint known to be sharp. The characterization is statistical: obtaining a generally efficient method for solving the posterior stopped experiment remains a separate computational problem.

Relation to prior work. For the stationary stochastic model studied here, [Agarwal et al. \(2013\)](#) first obtained $\tilde{O}(\text{poly}(d)\sqrt{T})$ regret; later localization and second-order methods progressively improved the dimension dependence ([Lattimore and György, 2021a, 2023](#); [Fokkema et al., 2024](#)). Information ratios and DEC formalize one-interaction regret–information or decision–estimation comparisons and turn them into sequential guarantees by re-optimization, Cauchy–Schwarz, testing, and information accumulation ([Russo and Van Roy, 2016](#); [Lattimore and György, 2021b](#); [Foster et al., 2021, 2023](#)). AIR contributes the exact posterior-coordinate identity used here ([Xu and Zeevi, 2025](#)); related work connects Bayesian information ratios to frequentist DEC ([Neu et al., 2024](#)). Chained ratios can also organize information across action scales, giving the tight linear-bandit rate with fixed two-query updates ([Gouverneur et al., 2024](#)). Our local optimization instead takes the ratio only after accumulating cost and progress over a variable-length experiment. The dimple theorem separates these local experiment classes; it does not show that sequential AIR or DEC algorithms fail.

Classical sequential design jointly optimizes adaptive experiments and stopping ([Chernoff, 1959](#); [Naghshvar and Javidi, 2013](#)); its objective is terminal evidence rather than BCO regret. Likewise, simple-regret work relates terminal recommendation and cumulative regret ([Bubeck et al., 2011](#)), and [Hanneke and Wang \(2025\)](#) characterize adaptive PAC query complexity for stochastic noisy bandits. The present first-hit object is posterior-local and charges both sample count and cumulative exploratory loss before the next risk contraction.

Interactive Fano is multistep from the outset because it compares complete adaptive transcript laws ([Chen et al., 2024](#)); the estimator-dependent ghost-mass lower envelope of [Xu \(2026c\)](#) is a one-scale specialization. Proposition 2.4 gives the bridge needed here: a Bellman cost-to-information bound and a Fano lower bound on the entropy required for posterior-risk contraction together prove a local premise $\Pi_b \geq \lambda$. The first-hit converse then performs the separate BCO-specific task of converting that premise into regret.

Near-optimal targets and localization arise in rate–distortion theory, satisficing, and scale-sensitive ratio analyses ([Dong and Van Roy, 2018](#); [Russo and Van Roy, 2022](#); [Arumugam and Van Roy, 2021](#); [Bubeck and Sellke, 2023](#)). Modern approaches to data-dependent localization also include pointwise notions of complexity ([Li and Xu, 2026](#); [Xu, 2026c,b](#)). Polynomial gains from measurements that orient later probes also arise in best-arm identification ([Maiti et al., 2026](#)). The exact-posterior state/index recursion and Bellman–Fano certificates of [Xu \(2026a\)](#) provide the general template. The present paper specializes it through the BCO cost $b^2\mathbb{E}N + b\mathbb{E}R$, a

retained sublevel-set target, risk-scale gluing, and a separate BCO-specific first-hit converse. Finally, [Lattimore and Szepesvári \(2019\)](#) gave a measure-theoretic minimax theorem for finite-action partial monitoring with general signal spaces. Our direct Gaussian argument handles the continuous action space used here; it is supporting quantifier closure rather than the source of the rate characterization.

Closest to the present feedback and loss model, [Rajaraman \(2026\)](#) prove a $\tilde{\Omega}(d^{5/4}\sqrt{T})$ regret lower bound against arbitrary observation-adaptive algorithms. Combined with Theorem 2.8, their theorem implies the class-level constraint $\alpha_{\text{stop}} \geq 3/2$. This implication is class-level; identifying a posterior state that directly witnesses the constraint remains open. By contrast, [Zhang et al. \(2026\)](#) obtain fully adaptive lower bounds for noiseless terminal-accuracy convex optimization. Their posterior-stability mechanism may inform direct profile lower bounds, but exact queries without cumulative loss define a different problem from the Gaussian regret model studied here.

Section 2 states the two-sided principle and its exact exponent consequence. Section 3 compares the dimple separation with the full-class profile and isolates the remaining question. Section 4 separates the reusable stationary template from its BCO-specific ingredients. The appendices prove the stochastic transfer and calibrations, the dimple separation, and the measurable and minimax completion steps.

2 The two-sided stopped-experiment principle

Formally, a loss-class sequence $\mathfrak{F} = (\mathfrak{F}_d(C))$ assigns to each dimension d and each known compact convex body $C \subset \mathbb{R}^d$ with nonempty interior a Borel class of continuous $[0, 1]$ -valued convex losses. Suprema over C below range over these bodies, and $\Pi_T(C)$ denotes the randomized nonanticipating Borel policies on C for horizon T .

We first state the main result. Fix $0 < c_A < \gamma < \vartheta < 1$. At risk scale b , use the persistent target

$$A_b^\circ \mid (F = f) \sim \text{Unif}\{x \in C : f(x) - \min_C f \leq c_A b\}. \quad (12)$$

Its construction and information bound are proved in Lemma A.1. A state $s = (Q, b)$ is active when $\vartheta b < r_B(Q) \leq b$. An admissible experiment uses a positive, integrable number of queries and stops no later than the first posterior with Bayes risk at most γb . If Z is its transcript, N_Z its length, R_Z its exploratory regret, and $\mathbf{M}$ the lower-scale hit, put

$$\rho(s, \mathcal{E}) = \frac{b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z}{I_s(A_b^\circ; Z) + d\mathbb{P}_s(\mathbf{M})}. \quad (13)$$

The information-assisted ratio ρ is the default. Its hit-only ablation is

$$\rho_{\text{hit}}(s, \mathcal{E}) = \frac{b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z}{d\mathbb{P}_s(\mathbf{M})}, \quad (14)$$

with value $+\infty$ when the denominator vanishes. It does not depend on the auxiliary target.

Proposition 2.1 (Aggregate information-ratio identity). *Fix an active state $s = (Q, b)$ and an admissible experiment $\mathcal{E}$. For every executed query, let*

$$\ell_t(H_{t-1}) = \mathbb{E}_s[F(X_t) - F_C^* \mid H_{t-1}], \quad i_t(H_{t-1}) = I_s(A_b^\circ; X_t, O_t \mid H_{t-1}),$$

including the randomized query law. Then

$$\mathbb{E}_s R_Z = \mathbb{E}_s \sum_{t \leq N_Z} \ell_t, \quad I_s(A_b^\circ; Z) = \mathbb{E}_s \sum_{t \leq N_Z} i_t. \quad (15)$$

Put

$$n = \mathbb{E}_s N_Z, \quad r = \mathbb{E}_s R_Z, \quad g = I_s(A_b^\circ; Z) + d\mathbb{P}_s(\mathbf{M}), \quad \bar{\ell} = r/n.$$

If $g > 0$, then

$$\rho(s, \mathcal{E}) = \underbrace{\frac{r^2}{ng}}_{\Gamma_{\text{block}}(s, \mathcal{E})} \left\{ \left(\frac{b}{\bar{\ell}} \right)^2 + \frac{b}{\bar{\ell}} \right\}. \quad (16)$$

Every executed query is selected from a pre-query history before the first hit, so $\bar{\ell} \geq \gamma b$. Consequently, on scale-calibrated experiments satisfying $\bar{\ell} \leq Kb$, one has $\rho(s, \mathcal{E}) \asymp_{\gamma, K} \Gamma_{\text{block}}(s, \mathcal{E})$. If $\mathbf{M}$ is empty, then $g = I_s(A_b^\circ; Z)$; for a one-query experiment, Γ_{block} is exactly the one-query information ratio for the retained near-optimal target A_b° .

Moreover, if the conditional one-query inequality $\ell_t(H_{t-1})^2 \leq G i_t(H_{t-1})$ holds almost surely on $\{t \leq N_Z\}$ at every pre-hit state reached by $\mathcal{E}$, then

$$\Gamma_{\text{block}}(s, \mathcal{E}) \leq G, \quad \rho(s, \mathcal{E}) \leq (\gamma^{-2} + \gamma^{-1})G. \quad (17)$$

Proof. The first identity in (15) is the tower property, and the second is the stopped chain rule. The algebraic identity (16) follows from $r = n\bar{\ell}$. Every executed query is selected from a pre-query history at which the first-hit condition has not yet occurred. The posterior Bayes risk there exceeds γb , and the conditional regret of any randomized query law is at least that risk, proving $r \geq \gamma bn$. The scale-calibrated comparison follows. Finally, Cauchy–Schwarz on the finite measure $\mu(dw, dt) = \mathbb{P}_s(dw)\mathbf{1}\{t \leq N_Z(\omega)\}$, whose mass is n , gives

$$r^2 \leq n\mathbb{E}_s \sum_{t \leq N_Z} \ell_t^2 \leq Gn\mathbb{E}_s \sum_{t \leq N_Z} i_t \leq Gng,$$

which proves the first inequality in (17). Since $b/\bar{\ell} \leq \gamma^{-1}$, the second follows from (16). $\square$

Proposition 2.1 makes the aggregate information-ratio interpretation precise. After the ratio–offset transformation, capped continuation is a Bellman optimal-stopping extension of the corresponding one-query offset family, augmented by terminal first-hit credit; the unrestricted ratio is the truncation limit of these capped values. The chain rule embeds any uniform one-step certificate, while allowing the posterior produced by early queries to determine the geometry of later ones.

For a known body C , let $S_{\mathfrak{F}, C}(d, B)$ be the active states with $B^{-1} \leq b \leq 1$; it is standard Borel because posterior risk is Borel. Define the *statewise profile* and the *measurable-policy profile* by

$$\Lambda_{\mathfrak{F}, C}^{\text{state}}(d, B) = \sup_{s \in S_{\mathfrak{F}, C}(d, B)} \inf_{\mathcal{E}} \rho(s, \mathcal{E}), \quad (18)$$

$$\Lambda_{\mathfrak{F}, C}^{\text{policy}}(d, B) = \inf_{\sigma_C} \sup_{\text{universally measurable } s \in S_{\mathfrak{F}, C}(d, B)} \rho(s, \sigma_C(s)), \quad (19)$$

Define $\Lambda_{\text{hit}, \mathfrak{F}, C}^{\text{state}}$ and $\Lambda_{\text{hit}, \mathfrak{F}, C}^{\text{policy}}$ by the same two displays with ρ_{hit} in place of ρ . Pointwise, $\rho \leq \rho_{\text{hit}}$, and hence

$$\Lambda_{\mathfrak{F}, C}^\diamond(d, B) \leq \Lambda_{\text{hit}, \mathfrak{F}, C}^\diamond(d, B), \quad \diamond \in \{\text{state, policy}\}. \quad (20)$$

Here σ_C is one universally measurable block policy on the augmented belief state: from the initial state s and each subsequent local history, it chooses the next action or stops; $\sigma_C(s)$ denotes the experiment that it induces from s . Since C is known, write

$$\Lambda_{\mathfrak{F}}^\diamond(d, B) := \sup_C \Lambda_{\mathfrak{F}, C}^\diamond(d, B), \quad \diamond \in \{\text{state, policy}\}. \quad (21)$$

Use the analogous notation $\Lambda_{\text{hit}, \mathfrak{F}}^\diamond$ for the bodywise hit-only profiles. Always $\Lambda^{\text{state}} \leq \Lambda^{\text{policy}}$.

For $\diamond \in \{\text{state}, \text{policy}\}$, define

$$\alpha_{\text{stop}}^\diamond(\mathfrak{F}) = \sup_{m \in \mathbb{N}} \limsup_{d \rightarrow \infty} \frac{\log\{1 + \Lambda_{\mathfrak{F}}^\diamond(d, d^m)\}}{\log d}. \quad (22)$$

Define $\alpha_{\text{hit}}^\diamond(\mathfrak{F})$ by replacing $\Lambda_{\mathfrak{F}}^\diamond$ in (22) with $\Lambda_{\text{hit}, \mathfrak{F}}^\diamond$. Let

$$\mathfrak{R}_{\mathfrak{F}}^{\text{M}}(d, T) := \sup_C \inf_{\pi \in \Pi_T(C)} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f R_T, \quad (23)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) := \sup_C \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi \in \Pi_T(C)} \mathbb{E}_Q R_T, \quad (24)$$

where the expectations include the Gaussian noise and the learner's randomization.

For $\diamond \in \{\text{B}, \text{M}\}$, define

$$\beta_{\mathfrak{F}}^\diamond := \sup_{M \in \mathbb{N}} \limsup_{d \rightarrow \infty} \sup_{2 \leq T \leq d^M} \frac{\log\{1 + \mathfrak{R}_{\mathfrak{F}}^\diamond(d, T)/\sqrt{T}\}}{\log d}. \quad (25)$$

Theorem 2.2 (Two-sided stopped-complexity transfer). *For every loss-class sequence $\mathfrak{F}$, every $T \geq 2$, and every $T^{-1} \leq \eta < 1/2$, there is $C = C_{\gamma, \vartheta, c_A} < \infty$ such that*

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, C\{1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)\} \frac{d\{1 + \log(1/\eta)\}}{\eta} + 2T\eta \right\}. \quad (26)$$

In particular,

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, \tilde{O} \left(\sqrt{d\{1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)\}T} \right) \right\}. \quad (27)$$

Conversely, for every $0 < \lambda < \Lambda_{\mathfrak{F}}^{\text{state}}(d, B)$, some active initial prior and scale $B^{-1} \leq b \leq 1$ satisfy, for every $n \geq 1$ and $0 < a < \gamma$,

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq \min \left\{ abn, \left[\frac{\lambda d(1 - a/\gamma)}{b} - bn \right]_+ \right\}. \quad (28)$$

In particular, whenever $d\lambda/b^2 \geq 8$,

$$n = \left\lfloor \frac{d\lambda}{4b^2} \right\rfloor, \quad (29)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq c_\gamma \sqrt{d\lambda n}. \quad (30)$$

Consequently, if $\liminf_{d \rightarrow \infty} \Lambda_{\mathfrak{F}}^{\text{state}}(d, 2) > 0$, then

$$\frac{1 + \alpha_{\text{stop}}^{\text{state}}}{2} \leq \beta_{\mathfrak{F}}^{\text{B}} \leq \frac{1 + \alpha_{\text{stop}}^{\text{policy}}}{2}. \quad (31)$$

Proposition 2.3 (First-hit profile and information ablation). *For every loss-class sequence, $T \geq 2$, and $T^{-1} \leq \eta < 1/2$,*

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, C_{\gamma, \vartheta} \{1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)\} \frac{d}{\eta} + 2T\eta \right\}. \quad (32)$$

Consequently,

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, O_{\gamma, \vartheta} \left(\sqrt{d \{1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)\} T} \right) \right\}. \quad (33)$$

Conversely, for every $0 < \lambda < \Lambda_{\text{hit}, \mathfrak{F}}^{\text{state}}(d, B)$, the all-horizon and calibrated lower bounds (28)–(30) hold for some active initial prior and scale. Hence, if the hit-only state profile is nondegenerate,

$$\frac{1 + \alpha_{\text{hit}}^{\text{state}}}{2} \leq \beta_{\mathfrak{F}}^{\text{B}} \leq \frac{1 + \alpha_{\text{hit}}^{\text{policy}}}{2}. \quad (34)$$

Proof. Fix a prior and select a block rule with hit-only ratio at most $1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)$. For block k in bucket j , with conditional hit probability p_k , the defining inequality is

$$b_j^2 \mathbb{E}[N_k \mid H_{S_k}] + b_j \mathbb{E}[R_k \mid H_{S_k}] \leq \{1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)\} dp_k.$$

The tower property identifies $\mathbb{E} \sum_{k:j(k)=j} p_k$ with the expected number of hits in that bucket, and Lemma A.6 bounds it by $(1 - \gamma/\vartheta)^{-1}$, including the horizon-crossing block. Therefore the expected exploratory regret is at most

$$C_{\gamma, \vartheta} \{1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)\} d \sum_{b_j \geq 2\eta} b_j^{-1} \leq C_{\gamma, \vartheta} \{1 + \Lambda_{\text{hit}, \mathfrak{F}}^{\text{policy}}(d, T)\} \frac{d}{\eta}.$$

Playing a posterior Bayes action below scale 2η adds at most $2T\eta$. Prior-wise Borelization by Lemma C.1, followed by the worst-prior supremum, proves (32); optimizing η proves (33).

For the converse, let the witnessing state satisfy $\Pi_b^{\text{hit}} \geq \lambda$. The policy-to-experiment construction in Proposition A.8 has $\mathbb{P}(\mathbf{M}) \geq 1 - a/\gamma$, so the hit-only defining inequality directly gives

$$b^2 n + b \text{BReg}_n \geq b^2 \mathbb{E} N_Z + b \mathbb{E} R_Z \geq \lambda d \mathbb{P}(\mathbf{M}) \geq \lambda d (1 - a/\gamma).$$

This is exactly the all-horizon lower inequality used there. Finally, Proposition A.12 constructs a constant-hit experiment, so on inverse-polynomial scales $\Lambda_{\text{hit}, \mathfrak{F}}^{\text{state}} \leq d^{2+o(1)}$ for every bounded convex subclass. The calibrated horizon is therefore polynomial, and the same dimension-growth calculation as in Theorem 2.2 gives (34). $\square$

The upper direction rests on one scale-amortization bound. Retain one target A_j across all stopped blocks Z_k begun at scale b_j , and let $p_k = \mathbb{P}(\mathbf{M}_k \mid H_k)$. Then

$$\mathbb{E} \sum_{k:j(k)=j} \{I(A_j; Z_k \mid H_k) + dp_k\} \leq d \log \frac{1}{c_A b_j} + \frac{d}{1 - \gamma/\vartheta}. \quad (35)$$

The information terms telescope because the same private target is retained. After a successful risk reduction, the posterior loss of the returned action is a nonnegative martingale starting below γb_j ; returning to the same active bucket requires it to exceed ϑb_j , so optional stopping bounds the return probability by γ/ϑ . Multiplying (35) by the scale costs b_j^{-1} and b_j^{-2} and summing geometrically proves the upper transfer. For the converse, convex averaging turns cumulative regret R_n into a terminal action of expected loss at most R_n/n ; stopping the policy at its first posterior-risk reduction then produces an admissible experiment and yields (28). The complete proof, including horizon truncation and monotone-convergence bookkeeping, is in Appendix A.

How Bellman continuation and lower-bound certification enter. Fix an active state, a query cap L , and a candidate ratio level $q > 0$. Proposition 2.1 identifies both the cost and target information as sums of conditional one-query increments. Write

$$c_s(\mathcal{E}) = b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z, \quad g_s(\mathcal{E}) = I_s(A_b^\circ; Z) + d\mathbb{P}_s(\mathbf{M}),$$

define $v_L(s) = \inf_{\mathcal{E}: N_Z \leq L} \rho(s, \mathcal{E})$, and put

$$J_L^q(s) := \inf_{\mathcal{E}: N_Z \leq L} \{c_s(\mathcal{E}) - qg_s(\mathcal{E})\} = \inf_{\mathcal{E}: N_Z \leq L} \mathbb{E}_s \left[\sum_{t \leq N_Z} \{b^2 + b\ell_t - qi_t\} - qd\mathbf{1}_M \right]. \quad (36)$$

Then

$$v_L(s) < q \iff J_L^q(s) < 0, \quad v_L(s) \geq q \iff J_L^q(s) \geq 0. \quad (37)$$

Increasing caps recover $\Pi_b(Q; A_b^\circ) = \inf_{L \geq 1} v_L(s)$, and therefore

$$\Pi_b(Q; A_b^\circ) < q \iff J_L^q(s) < 0 \text{ for some } L, \quad \Pi_b(Q; A_b^\circ) \geq q \iff J_L^q(s) \geq 0 \text{ for every } L. \quad (38)$$

The value in (36) obeys an information-penalized Bellman optimal-stopping recursion; the hit-only recursion is obtained by setting $i_t = 0$. Thus a negative value constructs a low-ratio experiment, while nonnegativity at every cap certifies a hard state. Cost and progress must be evaluated under the same experiment; they cannot be optimized separately. This is the precise multistep information-ratio interpretation: total cost is compared with total progress after continuation and stopping have been optimized. The global upper proof does not solve the original T -round Bayesian control problem. It glues these local stopped solutions through persistent-target information telescoping, return-time control, and geometric risk-scale balancing. Appendix C gives the recursion and the uncapped measurable closure.

The lower direction separates a local hardness certificate from its conversion into regret. Proposition 2.4 below gives one exact Bellman–Fano factorization of such a certificate. Once a local inequality $\Pi_b \geq \lambda$ is available, Proposition A.8 uses only the BCO-specific first-hit reduction: convex averaging and Markov’s inequality turn a low-regret policy into an experiment that reaches the lower risk scale with constant probability. No Fano comparison or Bellman supersolution is used in this transfer itself. This separation also explains why the same converse applies to the hit-only profile.

Proposition 2.4 (Bellman–Fano factorization of stopped hardness). *Fix an active initial state $s = (Q, b)$, and let A be any private target that is conditionally independent of Z given F for every admissible experiment. Write $\Pi_b(Q; A)$ and $\Pi_b^{\text{hit}}(Q)$ for the values in Definition A.4 at the empty history, and let $\mathcal{B}(Q, b)$ be the same class of admissible positive-length first-hit experiments used in those values. For a probability law ν on the same loss space $\mathfrak{F}_d(C)$ as Q , write*

$$r_B(\nu) := \inf_{x \in C} \int \{f(x) - \min_C f\} \nu(df).$$

Define the full-environment cost-to-information value and the Bayes-risk contraction entropy by

$$\Xi_b(Q) := \inf_{\mathcal{E} \in \mathcal{B}(Q, b)} \frac{b^2 \mathbb{E}_Q N_Z + b \mathbb{E}_Q R_Z}{I_Q(F; Z)}, \quad (39)$$

$$\mathcal{H}_b(Q) := \inf_{\substack{\nu \in \mathcal{P}(\mathfrak{F}_d(C)): \\ r_B(\nu) \leq \gamma b}} D_{\text{KL}}(\nu \| Q), \quad (40)$$

where a zero information denominator gives ratio $+\infty$. Activity and Pinsker's inequality imply $\mathcal{H}_b(Q) \geq 2(\vartheta - \gamma)^2 b^2 > 0$. If $\mathcal{H}_b(Q) < \infty$, then

$$\Pi_b(Q; A) \geq \Xi_b(Q) \frac{\mathcal{H}_b(Q)}{\mathcal{H}_b(Q) + d}, \quad \Pi_b^{\text{hit}}(Q) \geq \Xi_b(Q) \frac{\mathcal{H}_b(Q)}{d}. \quad (41)$$

For $\kappa > \gamma$, put

$$p_{b,\kappa}(Q) := \sup_{x \in C} Q\{F(x) - F_C^* \leq \kappa b\}. \quad (42)$$

Whenever $p_{b,\kappa}(Q) < 1 - \gamma/\kappa$,

$$\mathcal{H}_b(Q) \geq \text{kl}\left(1 - \frac{\gamma}{\kappa}, p_{b,\kappa}(Q)\right). \quad (43)$$

Proof. For every ν , $|r_B(\nu) - r_B(Q)| \leq \|\nu - Q\|_{\text{TV}}$ because the losses take values in $[0, 1]$. Thus every law feasible in (40) has total-variation distance at least $(\vartheta - \gamma)b$ from Q , and Pinsker's inequality proves the stated strict positivity of $\mathcal{H}_b(Q)$.

Let Q_Z be a regular conditional law of F given the complete stopped transcript. On $\mathbf{M}$, the posterior is feasible in (40). The posterior representation of mutual information therefore gives

$$I_Q(F; Z) = \mathbb{E}_Q D_{\text{KL}}(Q_Z \| Q) \geq \mathcal{H}_b(Q) \mathbb{P}_Q(\mathbf{M}). \quad (44)$$

If $I_Q(F; Z) = 0$, this display gives $\mathbb{P}_Q(\mathbf{M}) = 0$, and data processing gives $I_Q(A; Z) = 0$. The stopped denominator is then zero and both claimed ratio inequalities are immediate. Otherwise, data processing gives $I_Q(A; Z) \leq I_Q(F; Z)$, while the definition of $\Xi_b(Q)$ gives

$$b^2 \mathbb{E}_Q N_Z + b \mathbb{E}_Q R_Z \geq \Xi_b(Q) I_Q(F; Z).$$

Hence

$$I_Q(A; Z) + d \mathbb{P}_Q(\mathbf{M}) \leq I_Q(F; Z) \left\{1 + \frac{d}{\mathcal{H}_b(Q)}\right\}.$$

Dividing proves the first inequality in (41); using only $d \mathbb{P}_Q(\mathbf{M}) \leq d I_Q(F; Z) / \mathcal{H}_b(Q)$ proves the second.

For (43), fix a law ν with $r_B(\nu) \leq \gamma b$ and let x_ν be a Bayes action. Markov's inequality gives

$$\nu\{F(x_\nu) - F_C^* \leq \kappa b\} \geq 1 - \frac{\gamma}{\kappa},$$

whereas the Q -probability of the same set is at most $p_{b,\kappa}(Q)$. Put

$$q_\nu := \nu\{F(x_\nu) - F_C^* \leq \kappa b\}, \quad p_\nu := Q\{F(x_\nu) - F_C^* \leq \kappa b\}.$$

Then $p_\nu \leq p_{b,\kappa}(Q) < 1 - \gamma/\kappa \leq q_\nu$. Binary data processing and monotonicity of binary relative entropy on this region give

$$D_{\text{KL}}(\nu \| Q) \geq \text{kl}(q_\nu, p_\nu) \geq \text{kl}\left(1 - \frac{\gamma}{\kappa}, p_{b,\kappa}(Q)\right).$$

Taking the infimum over ν proves the claim. $\square$

The Bellman and Fano roles in Proposition 2.4 are separate and explicit. For a cap L and a proposed cost per unit information q , the max-form offset

$$\mathfrak{C}_L^{q,F}(s) := \sup_{\mathcal{E}: N_Z \leq L} \{q I_Q(F; Z) - b^2 \mathbb{E}_Q N_Z - b \mathbb{E}_Q R_Z\} \quad (45)$$

has an exact controlled Bellman recursion. Let $\mu' = \Psi(\mu, x, o)$ be the posterior after one query, let $i_F(\mu, x) = I_\mu(F; O \mid X = x)$, let $\ell(\mu, x)$ be posterior expected regret, and let $P_\mu(\cdot \mid x)$ be the predictive observation law. Set

$$\begin{aligned} \mathfrak{J}_k^{q,F}(\mu, b) &= \sup_{x \in C} \left\{ qi_F(\mu, x) - b^2 - b\ell(\mu, x) \right. \\ &\quad \left. + \int \mathbf{1}\{r_B(\mu') > \gamma b\} \mathfrak{U}_{k-1}^{q,F}(\mu', b) P_\mu(do \mid x) \right\}, \\ \mathfrak{U}_0^{q,F} &= 0, \quad \mathfrak{U}_k^{q,F} = \max\{0, \mathfrak{J}_k^{q,F}\}. \end{aligned} \tag{46}$$

The stopped chain rule and induction give $\mathfrak{C}_L^{q,F}(s) = \mathfrak{J}_L^{q,F}(Q, b)$. Because an admissible experiment has positive length, the root is $\mathfrak{J}_L^{q,F}$ and must execute one query; voluntary stopping through $\mathfrak{U}_k^{q,F}$ is available only afterward. The condition $\Xi_b(Q) \geq q$ is therefore equivalent to $\mathfrak{C}_L^{q,F}(s) \leq 0$ for every L .

For the first-hit truncations,

$$I_Q(F; Z^{[L]}) \uparrow I_Q(F; Z), \quad b^2 \mathbb{E}_Q N_{Z^{[L]}} + b \mathbb{E}_Q R_{Z^{[L]}} \uparrow b^2 \mathbb{E}_Q N_Z + b \mathbb{E}_Q R_Z.$$

Moreover, bounded Gaussian feedback gives $I_Q(F; Z) \leq \frac{1}{2} \log(5/4) \mathbb{E}_Q N_Z < \infty$. Thus the capped inequalities extend to every admissible integrable experiment. Bellman recursion prices information in experimental cost, while (43), or any ghost/small-ball certificate that uniformly lower-bounds $\mathcal{H}_b(Q)$, prices a risk contraction in information (Chen et al., 2024; Xu, 2026c,a). For the Rademacher linear prior of Proposition A.9, the proof gives $\Xi_b(Q_{\text{lin}}) \geq d$ and $\mathcal{H}_b(Q_{\text{lin}}) \geq d(1 - \gamma)^2/2$. Hence (41) recovers the order- d lower bound for both stopped profiles and explains why the terminal credit is normalized by d .

Remark 2.5 (Finite blockwise role of information). *Equality of the two profile exponents does not imply equality of their finite values or even of the credit assigned to a fixed experiment. For the Rademacher linear prior in Proposition A.9, make the single query $X = e_1$. The observation depends only on V_1 , so*

$$r_B(O) = b\{1 - \|\mathbb{E}[\theta_V \mid O]\| \} \geq b(1 - d^{-1/2}) > \gamma b$$

when $d > (1 - \gamma)^{-2}$. Thus $\mathbb{P}(\mathbf{M}) = 0$. The canonical sublevel-set target is uniform on a spherical cap and satisfies $\mathbb{E}[A_b^\circ \mid V] = \kappa \theta_V$ for some $\kappa > 0$. Since $O = b(1 - V_1/\sqrt{d}) + \xi$,

$$\text{Cov}((A_b^\circ)_1, O) = -\frac{\kappa b}{d} \neq 0,$$

and hence $I(A_b^\circ; O) > 0$.

The same phenomenon persists across genuinely multiple steps. Retain the same target and query $e_1, \dots, e_k$ in successive one-query blocks, where $k < (1 - \vartheta)^2 d$. After the first $j \leq k$ coordinates, the posterior mean of θ_V has norm at most $\sqrt{j/d} < 1 - \vartheta$, so the posterior Bayes risk remains above ϑb . Every intermediate posterior is therefore active at scale b , and none of these blocks hits the lower risk scale. Conditional on the preceding observations H_{j-1} , the unqueried sign V_j remains uniform and

$$\text{Cov}((A_b^\circ)_j, O_j \mid H_{j-1}) = -\frac{\kappa b}{d} \neq 0.$$

Hence each fresh coordinate observation has positive conditional information about the target, and these increments add by the chain rule. Thus information-assisted accounting composes partial progress across non-hitting blocks, whereas hit-only accounting assigns zero progress to each one. This separates the two blockwise accounting rules; it does not prove a strict separation between the optimized finite profiles or their resulting regret bounds.

2.1 Closing the quantifiers

Two regularity results close the remaining quantifiers in Theorem 2.2: Bayes–minimax duality identifies the global values, and measurable implementation assembles the statewise experiments into one admissible policy.

Proposition 2.6 (Bayes–minimax identity for Gaussian BCO). *For every finite d, T , every compact convex body C , and every Borel class of convex losses $\mathfrak{F}_d(C) \subset \mathcal{C}(C, [0, 1])$,*

$$\inf_{\pi \in \Pi_T(C)} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f R_T = \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi \in \Pi_T(C)} \mathbb{E}_Q R_T. \quad (47)$$

The supremum on the right is unchanged if it is restricted to finitely supported priors.

The next result shows that the two profiles in (18)–(19) have the same value in the models studied here.

Theorem 2.7 (Measurable implementation of statewise experiments). *Suppose C is compact and $\mathfrak{F}_d(C)$ is a Borel subset of $\mathcal{C}(C, [0, 1])$ with the uniform topology. Under Gaussian feedback, for every Borel target kernel into a standard Borel space and every $B < \infty$,*

$$\Lambda_{\mathfrak{F}, C}^{\text{state}}(d, B) = \Lambda_{\mathfrak{F}, C}^{\text{policy}}(d, B). \quad (48)$$

The same conclusion holds for the hit-only profiles:

$$\Lambda_{\text{hit}, \mathfrak{F}, C}^{\text{state}}(d, B) = \Lambda_{\text{hit}, \mathfrak{F}, C}^{\text{policy}}(d, B). \quad (49)$$

For each ratio and every $\epsilon > 0$, there is a universally measurable policy whose experiment has a finite cap depending on the state and whose ratio is within ϵ of the corresponding statewise infimum at every active state.

For a fixed prior and finite horizon, a universally measurable selector in Theorem 2.7 has a Borel version with the same induced strategic law; Lemma C.1 proves this prior-wise Borelization. Thus the selector is admissible in the Bayes value (24), whose policy class was defined using Borel kernels.

The full class of $[0, 1]$ -valued continuous convex losses is closed in $\mathcal{C}(C, [0, 1])$, so Theorem 2.7 applies. Write Λ and α_{stop} for the common information-assisted profile and exponent, Λ_{hit} and α_{hit} for their hit-only counterparts, and write $\mathfrak{R}_{\mathfrak{F}}$ and $\beta_{\mathfrak{F}}$ for the common Bayes–minimax value and dimension exponent.

Theorem 2.8 (Exact minimax exponent characterization). *Under the hypotheses of Theorem 2.7, if $\liminf_{d \rightarrow \infty} \Lambda_{\mathfrak{F}}(d, 2) > 0$, then*

$$\boxed{\beta_{\mathfrak{F}}^{\text{M}} = \beta_{\mathfrak{F}}^{\text{B}} = \frac{1 + \alpha_{\text{stop}}}{2} = \frac{1 + \alpha_{\text{hit}}}{2}.} \quad (50)$$

Thus the information-assisted profile is an exponent-preserving relaxation of first-hit complexity: both profiles characterize the same regret exponent, despite the different blockwise accounting exhibited in Remark 2.5.

Proposition 2.9 (Basic full-class calibration). *For the full bounded convex class and every $B \geq 2$,*

$$cd \leq \Lambda(d, B) \leq \Lambda_{\text{hit}}(d, B) \leq Cd^2 \text{polylog}(dB). \quad (51)$$

The fixed-scale display does not determine the polynomial-scale exponent; the current exponent bracket is recorded in (59) below.

2.2 Adaptive continuation on the dimple family

For $C = B_2^d(1)$, $\theta \in S_2^{d-1}(1)$, and $s = \sqrt{1 + 2\varepsilon}$, define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}(s - \|\theta - x\|)_+^2. \quad (52)$$

At the origin every member has the common loss

$$b_0 := F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1 + 2\varepsilon} + 1} \asymp \varepsilon. \quad (53)$$

These convex losses are uniquely minimized at θ and have minimum zero. We retain this loss family from Bakhtiari et al. (2025) and prove two new facts about it: fresh random directions remain costly even when their radii are adaptive, whereas observation-dependent directions yield a low-value stopped experiment. Let Π_b^{fresh} and $\Pi_b^{\text{fresh, hit}}$ denote the two ratios with the infimum restricted to policies that may adapt radii but draw a fresh uniform direction at every query, and let $\Pi_b^{[1], \text{hit}}$ denote the one-query restriction of Π_b^{hit} .

Theorem 2.10 (Adaptive continuation on the dimple family). *At $\varepsilon = d^{-1}$ and $b = b_0 \asymp d^{-1}$ in the dimple family (52), take the initial prior $\Theta \sim \text{Unif}(S_2^{d-1}(1))$ and use the persistent sublevel-set target (12). At this initial state, the one-query restriction and the class that may adapt radii but draws a fresh uniform direction at every query satisfy*

$$\Pi_b^{[1]} = \Omega_\gamma(d^2), \quad \Pi_b^{\text{fresh}} = \Omega_\gamma(d^2), \quad \Pi_b = \tilde{O}_\gamma(d). \quad (54)$$

Moreover,

$$\Pi_b^{[1], \text{hit}} = \Omega_\gamma(d^2), \quad \Pi_b^{\text{fresh, hit}} = \Omega_\gamma(d^2), \quad \Pi_b^{\text{hit}} = \tilde{O}_\gamma(d). \quad (55)$$

Thus observation-dependent direction selection reduces this local complexity by a factor $\tilde{\Omega}(d)$. Matching upper bounds for the two restricted classes are not claimed.

Corollary 2.11 (Minimax regret on the dimple family). *For $\varepsilon_d = d^{-1}$, let $\mathfrak{F}_d^{\text{dim}} = \{F_\theta : \theta \in S_2^{d-1}(1)\}$ on $B_2^d(1)$, and put*

$$L_{d,T} := 1 + \log\{C_\gamma d \max\{T, d\}[1 + \log\{d \max\{T, d\}\}]\}.$$

For all sufficiently large d and every $T \geq 2$,

$$\inf_{\pi \in \Pi_T(B_2^d(1))} \sup_{\theta \in S_2^{d-1}(1)} \mathbb{E}_\theta R_T \leq \min\{b_0 T, C_\gamma d \sqrt{T} L_{d,T}\} \leq C_\gamma L_{d,T} \min\left\{\frac{T}{d}, d\sqrt{T}\right\}. \quad (56)$$

This is an existential minimax bound; computational efficiency and a matching lower bound are not claimed. The adaptive term improves the origin strategy only beyond the crossover $T \gtrsim d^4 L_{d,T}^2$.

Proof. Every posterior Q on $\mathfrak{F}_d^{\text{dim}}$ satisfies $r_B(Q) \leq \mathbb{E}_Q F_\Theta(0) = b_0$. Hence no active state exists when $b \geq b_0/\gamma$. At every remaining active state, run Theorem B.2 at $a = b \wedge b_0$, for which $b/a \leq \gamma^{-1}$. Its every-prior guarantee, posterior-risk gate, and first-hit truncation give uniformly

$$\mathbb{P}_Q(\mathbf{M}) \geq c_\gamma, \quad b^2 \mathbb{E}_Q N_Z + b \mathbb{E}_Q R_Z \leq C_\gamma d^2 \mathcal{L}_{d,a}^2.$$

Since $b \geq T^{-1}$ and $b_0 \geq c/d$,

$$a = b \wedge b_0 \geq c \min\{T^{-1}, d^{-1}\}, \quad \mathcal{L}_{d,a} \leq C_\gamma L_{d,T}.$$

Hence the fixed-body hit-only state profile over scales $b \geq T^{-1}$ is at most $C_\gamma d L_{d,T}^2$. The map $\theta \mapsto F_\theta$ is continuous in the uniform norm, so $\mathfrak{F}_d^{\text{dim}}$ is compact and Borel; Theorem 2.7 gives the same measurable-policy profile. The fixed-body proof of Proposition 2.3, followed by Proposition 2.6, gives the second upper bound in (56). Playing the origin gives the first because $F_\theta(0) = b_0$ for every θ . $\square$

The stopped experiment has three stages. Here a is the target terminal loss and q is the current angular error:

stage	progress	(queries, regret)
orthogonal scan	find v_0 with $\langle \theta, v_0 \rangle = \tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$	$\tilde{O}(d^2/\varepsilon^2, d^2/\varepsilon)$
tangent amplification	use observation-adapted tangent differences to enter a fixed cone	$\tilde{O}(d^2/\varepsilon^2, d^2/\varepsilon)$
angular contraction	invert boundary observations and repeat $q \mapsto q/2$ until $q = O(a)$	$\tilde{O}(d^2/a^2, d^2/a)$

Only the last two stages choose later directions from earlier observations; this is precisely the adaptivity excluded by the fresh-direction lower bound.

3 What the characterization buys and what remains open

From a local profile to a dimensional advance. The characterization converts any improvement in the stopped profile directly into a regret improvement. If, for every fixed m ,

$$\Lambda^{\text{policy}}(d, d^m) \leq d^{\alpha+o_m(1)}, \quad (57)$$

then, for every fixed M , the upper transfer gives uniformly over $2 \leq T \leq d^M$,

$$\mathfrak{R}_{\mathfrak{F}}(d, T) \leq d^{(1+\alpha)/2+o_M(1)} \sqrt{T}. \quad (58)$$

Conversely, a statewise lower bound of order $d^{\alpha-o(1)}$ at an inverse-polynomial risk scale yields the corresponding obstruction at its calibrated polynomial horizon. Thus reducing a profile from d^2 to $d^{2-\kappa}$ improves the regret dimension exponent by $\kappa/2$. When the uniformly low-value experiments are explicit and implementable, the upper proof is also an algorithm-design principle. The unrestricted profile characterization alone remains statistical and does not provide an efficient solver for the posterior stopping problem. The dimple experiment is a concrete family-level instance of this conversion.

Consequence of current regret bounds. For the full bounded convex class, the best currently known regret bounds, combined with Theorem 2.8, give

$$\boxed{\frac{5}{4} \leq \beta_{\mathfrak{F}} \leq \frac{3}{2}}, \quad \boxed{\frac{3}{2} \leq \alpha_{\text{stop}} = \alpha_{\text{hit}} \leq 2}. \quad (59)$$

The lower endpoint follows from the independent theorem of Rajaraman (2026). To embed its 1-Lipschitz hard family in the present bounded model, set $g = (f + 1)/2$ and, from $Y = f(x) + \xi$, return $(Y + 1)/2 + \zeta$, where $\zeta \sim N(0, 3/4)$ is independent; the result is $g(x) + \xi'$ with $\xi' \sim N(0, 1)$. The best known stochastic upper bound supplies the other endpoint. Neither endpoint is known to be sharp.

What the paper’s examples establish. The following calibrations are proved here:

family	stopped conclusion	interpretation
linear losses	order- d profile	recovers the linear-bandit dimension
well-conditioned quadratics	order- d profile	unknown center and orientation do not create a superlinear profile
dimple family	one-query and fresh-direction values $\Omega(d^2)$, but unrestricted stopped value $\tilde{O}(d)$ for both profiles	observation-dependent directions remove a polynomial one-step obstruction and yield Corollary 2.11

The first two rows show that unknown orientation alone does not create the full-class gap. The third shows that weak directional information can become easy after it is used to orient later probes. None of these statements determines the full-class exponent.

Open question 3.1 (Full-class stopped exponent). *Determine $\alpha_{\text{stop}} = \alpha_{\text{hit}}$ for the full bounded convex class. Equivalently, by Theorem 2.8, determine its minimax dimension exponent $\beta_{\mathfrak{F}}$. Any improvement to either side of (59) must come from a corresponding upper or lower control of an unrestricted stopped profile.*

The profile is invariant under invertible affine changes of variables: such a map transports the target and every stopped experiment while preserving observations, regret, information, and the first-hit event. For every fixed admissible triple (c_A, γ, ϑ) , the exact characterization also identifies both stopped exponents with $2\beta_{\mathfrak{F}} - 1$; thus the exponent, though not necessarily either finite profile value, is invariant under these threshold constants. These facts support a geometric interpretation. We nevertheless avoid calling the quantity an intrinsic dimension because invariance under alternative target kernels has not been established.

4 Scope beyond BCO

The portable principle is local-to-global. At each posterior state, define a persistent target or a first-hit event, optimize accumulated experimental cost against accumulated progress over an adaptive stopped block, and then amortize those blocks across risk scales. A reverse first-hit reduction tests whether the resulting local quantity is also necessary. This template may extend to fixed-environment stochastic bandits, noisy optimization, and fixed-law partial monitoring, but each model requires its own target, terminal progress, amortization, and measurability arguments.

For Gaussian BCO, convex averaging, the sublevel-set target, Gaussian selection, and the cost $b^2\mathbb{E}N + b\mathbb{E}R$ are problem-specific. At a fixed state, the ratio–offset form is a Bellman optimal-stopping problem; see (114)–(117). Globally, the proof combines its local solutions rather than solving the original T -round Bellman equation. Computational tractability therefore requires additional structure.

The equality of exponents gives a two-part interpretation. Adaptive continuation until the first risk contraction determines the polynomial dimension exponent, while mutual information supplies chain-rule-additive accounting for partial progress along the way. The dimple separation shows that passing from one query to an observation-adaptive stopped experiment can change the relevant dimension by a polynomial factor. For the full convex class, the remaining question is whether unrestricted continuation can achieve a profile $d^{2-\kappa}$ for some $\kappa > 0$; one-query coefficients alone cannot settle this question.

A Proofs for the two-sided stopped-experiment principle

A.1 An affine-covariant near-optimal target

Assume that every loss in the prior has range at most $0 < \Delta \leq 1$ on C . The analysis needs a near-optimal random index whose information can telescope without depending on an arbitrary choice among multiple minimizers. For $0 < \eta < \Delta$, put

$$S_\eta(f) = \{x \in C : f(x) - \min_C f \leq \eta\}, \quad A_\eta^\circ \mid (F = f) \sim \text{Unif}(S_\eta(f)). \quad (60)$$

The conditional draw uses private randomness unavailable to the policy. Near-optimal random learning targets are standard in rate-distortion and satisficing analyses (Dong and Van Roy, 2018; Arumugam and Van Roy, 2021; Russo and Van Roy, 2022). The role here is more specific: the target is affine covariant, retained across repeated local experiments, and paired with a posterior-risk hitting event.

Lemma A.1 (Sublevel-set near-optimal target). *Equip the continuous losses on C with the uniform topology. The map $(f, \eta) \mapsto \text{Unif}(S_\eta(f))$ is a Borel probability kernel on the loss space times $(0, \Delta)$. In particular, for every fixed η , the coupling (60) satisfies*

$$F(A_\eta^\circ) - F_C^* \leq \eta \quad \text{almost surely}, \quad I(F; A_\eta^\circ) \leq d \log \frac{\Delta}{\eta}. \quad (61)$$

If T is an invertible affine map, then the sublevel-set target for $f \circ T^{-1}$ on $T(C)$ has the law of $T(A_\eta^\circ)$. Thus the channel is affine covariant and independent of a minimizer representation or of the prior on F .

Proof. The graph of the correspondence $(f, \eta) \mapsto S_\eta(f)$ is Borel because $f \mapsto \min_C f$ is continuous in the uniform norm and $(f, x) \mapsto f(x)$ is continuous. More explicitly, $(f, \eta, x) \mapsto \mathbf{1}\{x \in S_\eta(f)\}$ is a bounded jointly Borel function. Its integral against Lebesgue measure is therefore Borel in (f, η) , and the same parameter-integration argument, now treating (f, η) jointly as the parameter and multiplying by the indicator of an arbitrary Borel set in C , proves that $(f, \eta) \mapsto \text{Unif}(S_\eta(f))$ is a Borel probability kernel once positivity of the denominator is shown.

Fix any $y \in \arg \min_C f$ and put $\alpha = \eta/\Delta$. For every $x \in C$, convexity gives

$$f((1 - \alpha)y + \alpha x) \leq (1 - \alpha)f(y) + \alpha f(x) \leq f_C^* + \eta.$$

Consequently, $(1 - \alpha)y + \alpha C \subseteq S_\eta(f)$ and $\text{vol}(S_\eta(f)) \geq \alpha^d \text{vol}(C) > 0$. The support definition gives the near-optimality statement. Using the uniform law on C as a reference,

$$I(F; A_\eta^\circ) \leq \mathbb{E} D_{\text{KL}}(P_{A_\eta^\circ \mid F} \parallel \text{Unif}(C)) = \mathbb{E} \log \frac{\text{vol}(C)}{\text{vol}(S_\eta(F))} \leq d \log(1/\alpha).$$

Finally, invertible affine maps carry sublevel sets to sublevel sets and uniform measures to uniform measures, proving covariance. $\square$

For the reduction, fix a tolerance η and write $A = A_\eta^\circ$. On an auxiliary probability space, conditionally on F , draw the countable family of bucket targets $A_j \sim A_{c_A b_j}^\circ$, for a fixed $c_A \in (0, 1)$, before the interaction begins, using independent private coins. The learner never observes these coins. Target A_j is activated for the analysis when bucket j is first entered and is retained across every later return to that bucket. This ex-ante coupling lets its information drift telescope through the random bucket-entry history.

Let h be a reached history and write

$$r_B(h) = \inf_{x \in C} \mathbb{E}[F(x) - F_C^* \mid h], \quad \Phi(h) = I(F; A \mid h). \quad (62)$$

Throughout, conditional information at a random history is interpreted statewise: $I(A; Z \mid H)$ denotes the H -measurable function which, at $H = h$, equals $D_{\text{KL}}(P_{A,Z|h} \| P_{A|h} \otimes P_{Z|h})$. An admissible experiment has a complete, almost surely finite stopped transcript Z , including its length, actions, observations, and private coins. Its actions cannot use the hidden auxiliary randomization in A . This operational restriction ensures the Markov relation

$$A \perp Z \mid (F, h). \quad (63)$$

In particular, after every history generated without access to the target's private coin, the conditional law of A given F remains its original kernel $K_b(\cdot \mid F)$.

Lemma A.2 (Exact information drift). *Every admissible experiment satisfies*

$$\Phi(h) - \mathbb{E}[\Phi(h, Z) \mid h] = I(A; Z \mid h). \quad (64)$$

Consequently, along any adaptive sequence of admissible blocks,

$$\mathbb{E} \sum_k I(A; Z_k \mid H_k) \leq I(F; A). \quad (65)$$

Proof. Expand $I(A; F, Z \mid h)$ in two orders and use $I(A; Z \mid F, h) = 0$:

$$I(A; F \mid h) = I(A; Z \mid h) + \mathbb{E}[I(A; F \mid h, Z) \mid h].$$

Iterating this identity proves (65); the remaining conditional mutual information is nonnegative. $\square$

The information term $I(A; Z \mid h)$ is the posterior average of the coordinate log gain underlying AIR (Xu and Zeevi, 2025). Under the Markov relation (63), the potential $I(F; A \mid h)$ has exactly this fixed-index drift, as used by Xu (2026a). The BCO-specific step is to couple the drift to a near-optimal target and the first hitting time of a lower posterior-risk level.

A frequentist high-probability recommendation is not automatically a posterior guarantee: the bound must hold under the terminal posterior on every accepted transcript. The following elementary acceptance rule enforces this condition.

Lemma A.3 (Posterior-risk acceptance). *Suppose a stopped routine returns $\hat{x} = \hat{x}(Z)$ and*

$$\mathbb{E}[F(\hat{x}) - F_C^* \mid h] \leq ab.$$

For $\gamma > a$, accept the recommendation only on

$$\mathbf{M} = \{\mathbb{E}[F(\hat{x}) - F_C^* \mid h, Z] \leq \gamma b\}.$$

Then every accepted transcript has terminal posterior regret at most γb , and

$$\mathbb{P}(\mathbf{M} \mid h) \geq 1 - a/\gamma.$$

Proof. The terminal posterior regret is nonnegative and has expectation at most ab by the tower property. Apply Markov's inequality. $\square$

The lemma validates accepted transcripts, but does not pay for rejected ones. A complete experiment must still charge its entire expected query and regret cost.

Whenever a capped routine below is analyzed through this rule, we couple it to its full scheduled run but execute its first-hit truncation: after each scalar observation, stop as soon as $r_B \leq \gamma b$ and return a posterior Bayes action. The gated terminal event implies that a hit occurred by the cap, while truncation only decreases nonnegative query and regret costs. Thus all explicit linear and dimple bounds below concern the first-hit complexity in Definition A.4.

A.2 Stopped experiment complexity

Fix constants

$$0 < \gamma < \vartheta < 1 \quad (66)$$

and call (h, b) *active* when

$$\vartheta b < r_B(h) \leq b. \quad (67)$$

For an experiment Z , let N_Z be its length and

$$R_Z = \sum_{t=1}^{N_Z} \{F(X_t) - F_C^*\}.$$

The experiment stops when its goal has already been reached. Starting from $H_0 = h$, let

$$\tau_b = \inf\{t \geq 1 : r_B(H_t) \leq \gamma b\}. \quad (68)$$

A *first-hit experiment* chooses a positive adapted stopping time σ , stops at $\tau = \sigma \wedge \tau_b$, and has a complete transcript $Z = H_\tau \setminus H_0$. It declares $\mathbf{M} = \{\tau_b \leq \sigma\}$ and, on $\mathbf{M}$, returns a measurable posterior Bayes action x_Z . Thus

$$\mathbb{E}[F(x_Z) - F_C^* \mid h, Z] = r_B(h, Z) \leq \gamma b \quad \text{on } \mathbf{M}. \quad (69)$$

On $\mathbf{M}^c$ the experiment stops before the posterior reaches the smaller scale. No pathwise charge is assigned to either event; the inequalities below control the entire experiment in expectation. The first-hit convention prevents an experiment from continuing after solution merely to harvest target information that the regret problem no longer needs.

The information and risk-reduction terms combine into one statewise quantity.

Definition A.4 (Information-assisted and hit-only stopped complexity). *Fix an active state (h, b, A) satisfying (67), where A is the persistent near-optimal target sampled once for this geometric scale. Let $\mathcal{B}(h, b, A)$ be the collection of admissible positive-length, almost surely finite first-hit experiments $(Z, \mathbf{M}, x_Z)$ described around (68), with $\mathbb{E}[N_Z \mid h] < \infty$. Define*

$$\Pi_b(h; A) = \inf_{(Z, \mathbf{M}, x_Z) \in \mathcal{B}(h, b, A)} \frac{b^2 \mathbb{E}[N_Z \mid h] + b \mathbb{E}[R_Z \mid h]}{I(A; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)}. \quad (70)$$

Its hit-only counterpart is

$$\Pi_b^{\text{hit}}(h) = \inf_{(Z, \mathbf{M}, x_Z) \in \mathcal{B}(h, b, A)} \frac{b^2 \mathbb{E}[N_Z \mid h] + b \mathbb{E}[R_Z \mid h]}{d\mathbb{P}(\mathbf{M} \mid h)}. \quad (71)$$

The collection in (71) is independent of the choice of A , so the target is omitted from the notation. Each ratio is $+\infty$ when its denominator is zero. The target's private randomization remains inaccessible to the policy, and the complete transcript includes the stopping time, actions, observations, and policy coins. All complexities below use this first-hit convention.

Definition A.4 is a stopped analogue of the one-round tradeoff measured by localized and constrained variants of the decision–estimation coefficient (Foster et al., 2021, 2023). It replaces one randomized query by a variable-length BCO experiment whose continuation rule and stopping time are optimized jointly. It also retains a near-optimal target across repeated experiments and counts the first hit of a lower Bayes-risk level as progress. The hit-only ratio retains just the second channel. The coordinate log identity underlying AIR supplies the telescoping algebra (Xu and Zeevi, 2025); its fixed-index Bellman extension (Xu, 2026a) supplies the information-assisted continuation recursion.

Proposition A.5 (One query and adaptive continuation). *Let $\mathcal{B}_1(h, b, A)$ be the subclass in Definition A.4 with $N_Z = 1$ almost surely, and let $\Pi_b^{[1]}(h; A)$ be the corresponding restricted infimum. Then*

$$\Pi_b(h; A) \leq \Pi_b^{[1]}(h; A). \quad (72)$$

For a one-query law ν whose posterior remains active almost surely (so $\mathbf{M} = \emptyset$), write

$$i_\nu = I(A; X, O \mid h), \quad \ell_\nu = \mathbb{E}[F(X) - F_C^* \mid h], \quad \Gamma_\nu = \frac{\ell_\nu^2}{i_\nu}. \quad (73)$$

When $i_\nu > 0$,

$$\frac{b^2 + b\ell_\nu}{i_\nu} = \Gamma_\nu \left\{ \left(\frac{b}{\ell_\nu} \right)^2 + \frac{b}{\ell_\nu} \right\}. \quad (74)$$

Hence, at a state satisfying (67), the two ratios are comparable up to constants on query laws with $\ell_\nu \leq Kb$, for fixed $K < \infty$. They need not be comparable for high-loss probes.

Assume $I(F; A \mid h) < \infty$. For $\lambda > 0$, define the one-query offset

$$\mathfrak{D}_{b,\lambda}^{[1]}(h; A) = \inf_{(Z, \mathbf{M}, x_Z) \in \mathcal{B}_1(h, b, A)} [b^2 + b\mathbb{E}[R_Z \mid h] - \lambda \{I(A; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)\}]. \quad (75)$$

Then

$$\Pi_b^{[1]}(h; A) \leq \lambda \iff \mathfrak{D}_{b,\lambda}^{[1]}(h; A) \leq 0. \quad (76)$$

For finite problems and experiments capped at m queries, iterating the same offset with its posterior continuation gives the standard information-penalized optimal-stopping recursion. After the ratio-offset transformation, the capped values are obtained by Bellman optimal-stopping continuation of the one-query offset family; by Proposition C.4, Π_b is their uncapped limit. Attainment by one infinite-horizon optimizer is not needed.

Proof. The class inclusion gives (72), and (74) is algebra. Under (67), every query law has $\ell_\nu \geq r_B(h) > \vartheta b$; together with $\ell_\nu \leq Kb$, this bounds the factor in braces above and below by positive constants depending only on K and ϑ . Finally, write $c(Z) = b^2 + b\mathbb{E}[R_Z]$ and $g(Z) = I(A; Z) + d\mathbb{P}(\mathbf{M})$. Data processing gives the uniform bound $g(Z) \leq I(F; A \mid h) + d < \infty$. Hence the infimum of $c(Z)/g(Z)$ is at most λ exactly when the infimum of $c(Z) - \lambda g(Z)$ is nonpositive, including the boundary case in which the ratio approaches λ without attaining it. The convention for $g(Z) = 0$ is harmless because $c(Z) \geq b^2 > 0$. This proves (76). $\square$

One-round coefficients and stopped experiments. For one randomized BCO query, the target-specific information ratio is ℓ_ν^2/i_ν . Equation (74) is the exact scale calibration, and Proposition 2.1 shows how these one-query increments embed in a stopped block. Offset DEC and constrained regret-DEC likewise optimize one randomized decision and its induced observation, balancing immediate loss against statistical separation from a reference model. Their associated algorithms may re-solve the one-round problem after every update. Here the local complexity itself performs that continuation: it optimizes the entire observation-adaptive query sequence and its stopping decision before total cost is divided by total progress. This is Bellman optimal-stopping continuation of the scale-calibrated one-query offset family, not a claim of general numerical equivalence among Π_b , DEC, and the information ratio.

A.3 Upper and lower transfers

Fix a prior Q . In this subsection write $\text{BReg}_T := \mathbb{E}_Q^\pi R_T$, and let N_{explore} be the number of actually executed rounds inside selected first-hit blocks; rounds on which a posterior Bayes action is played are not included.

Lemma A.6 (Stopped-block and horizon accounting). *Consider a finite-horizon policy that successively invokes positive-length, almost surely finite first-hit blocks. Let $S_k < T$ be the start of block k , $j(k)$ its risk bucket, $\bar{Z}_k$ its complete scheduled transcript, and*

$$p_k = \mathbb{P}(\mathbf{M}_k \mid H_{S_k}), \quad K_j = \sum_{k:j(k)=j} \mathbf{1}_{\mathbf{M}_k}.$$

If the horizon crosses the last block, complete that block virtually with fresh Gaussian observations and policy coins; the virtual observations are used only in the accounting. Then the executed prefix has no larger query or regret cost than $\bar{Z}_k$, and, for every bucket j ,

$$\mathbb{E} \sum_{k:j(k)=j} I(A_j; \bar{Z}_k \mid H_{S_k}) \leq I(F; A_j), \quad (77)$$

$$\mathbb{E} \sum_{k:j(k)=j} p_k = \mathbb{E} K_j \leq \frac{1}{1 - \gamma/\vartheta}. \quad (78)$$

The conditional information in (77) is the statewise quantity, followed by expectation over the random block-start history.

Proof. Enlarge the probability space by an independent array of Gaussian noise and policy coins. If one block straddles T , use this array to run its scheduled continuation after time T ; all earlier blocks are already complete. Since query count and regret are nonnegative, replacing the executed final prefix by its virtual completion can only increase both costs.

Fix j and first truncate every complete block after m queries. Insert the transcripts of all intervening blocks, posterior-Bayes rounds, and stopping markers. The ordinary finite chain rule gives

$$\mathbb{E} \sum_{k:j(k)=j} I(A_j; \bar{Z}_k^{[m]} \mid H_{S_k}) \leq I(A_j; H_{\text{virt}}^{[m]}) \leq I(A_j; F).$$

The first inequality discards nonnegative information increments outside bucket j . The second uses $A_j \perp H_{\text{virt}}^{[m]} \mid F$, because neither the actual nor virtual policy observes the target's private coin. Increasing transcript prefixes generate each complete stopped transcript, so relative-entropy monotone convergence proves (77).

The tower property gives $\mathbb{E} \sum_{k:j(k)=j} p_k = \mathbb{E} K_j$. After the m th hit in bucket j , let x_m be the returned action. Conditionally on that stopped history,

$$q_t = \mathbb{E}[F(x_m) - F_C^* \mid H_t]$$

is a bounded nonnegative martingale starting at most γb_j . A later return to bucket j is possible only at a stopping time when $r_B(H_t) > \vartheta b_j$, which forces $q_t > \vartheta b_j$. Doob's maximal inequality therefore gives

$$\mathbb{P}(K_j \geq m + 1 \mid K_j \geq m, \text{history at the } m\text{th hit}) \leq \gamma/\vartheta.$$

Thus $\mathbb{P}(K_j \geq m) \leq (\gamma/\vartheta)^{m-1}$, and summing the tail probabilities proves (78). A virtually completed last block can contribute one final hit but no later return, so the same tail argument includes it. $\square$

Theorem A.7 (From local experiments to global regret). *Fix $0 < c_A < \gamma$. Conditionally on F , draw independent private targets $A_j = A_{c_A b_j}^\circ$ for the geometric risk scales $b_j = \Delta \vartheta^j$ before interaction, and retain A_j across every return to scale j . Suppose that, at every active state, a universally measurable rule selects a first-hit experiment whose ratio in (70) is at most Λ_d , uniformly over scales and posterior states. Then, for every $0 < \eta < \Delta/2$,*

$$\text{BReg}_T \leq C_{\gamma, \vartheta, c_A} \Lambda_d \frac{d\{1 + \log(\Delta/\eta)\}}{\eta} + 2T\eta, \quad (79)$$

$$\mathbb{E}N_{\text{explore}} \leq C_{\gamma, \vartheta, c_A} \Lambda_d \frac{d\{1 + \log(\Delta/\eta)\}}{\eta^2}. \quad (80)$$

Consequently,

$$\text{BReg}_T = \tilde{O}(\sqrt{d\Lambda_d T}). \quad (81)$$

Thus $\Lambda_d = \tilde{O}(d^{2-\kappa})$ gives $\tilde{O}(d^{(3-\kappa)/2}\sqrt{T})$.

Proof. Invoke the selected experiment whenever $r_B > 2\eta$ and play a posterior Bayes action otherwise. The ratio bound implies, at scale b_j ,

$$\mathbb{E}[R_Z \mid h] \leq \frac{\Lambda_d}{b_j} \{I(A_j; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)\},$$

and the analogous query bound with b_j^{-2} . Lemma A.6 bounds the information sum in bucket j by $I(F; A_j)$ and its expected hit count by $(1 - \gamma/\vartheta)^{-1}$, including a virtually completed horizon-crossing block. Lemma A.1 gives $I(F; A_j) \leq d \log\{\Delta/(c_A b_j)\}$. Summing b_j^{-1} and b_j^{-2} over the geometric buckets down to scale 2η yields the displayed multiples of $d\{1 + \log(\Delta/\eta)\}/\eta$ and $d\{1 + \log(\Delta/\eta)\}/\eta^2$. Below that scale, a posterior Bayes action incurs conditional expected regret at most 2η per round. Combining these bounds proves (79)–(80); optimizing in η proves (81). $\square$

The same ratio turns a local bound that holds for every stopped experiment into a Bayes-regret lower bound. This converse is why the lower-bound direction of the core question is stronger than disproving one sufficient condition. For an initial law Q , write $\Pi_b(Q; A)$ for the stopped complexity at the empty history.

Proposition A.8 (From stopped complexity to a regret lower bound). *Suppose an initial prior, a target coupling fixed independently of the policy under test, an active scale b , and $\lambda > 0$ satisfy*

$$\Pi_b(Q; A) \geq \lambda. \quad (82)$$

Then, for every integer $n \geq 1$ and every $0 < a < \gamma$, every n -round adaptive policy satisfies

$$\text{BReg}_n \geq \min \left\{ abn, \left[\frac{\lambda d(1 - a/\gamma)}{b} - bn \right]_+ \right\}. \quad (83)$$

In particular, if $d\lambda/b^2 \geq 8$, then at

$$n = \left\lfloor \frac{d\lambda}{4b^2} \right\rfloor \quad (84)$$

every n -round adaptive policy has

$$\text{BReg}_n \geq c_\gamma \frac{d\lambda}{b} = \Omega_\gamma(\sqrt{d\lambda n}). \quad (85)$$

Because this is a Bayes lower bound for an initial prior, it immediately lower-bounds the minimax regret of every loss class containing $\text{supp}(Q)$. At a reached state the same argument gives only a conditional continuation bound unless the state is made initial or is reached with controlled probability.

Proof. Fix a policy and let $R_n = \mathbb{E} \sum_{t=1}^n \{F(X_t) - F_C^*\}$. If $R_n > abn$, the first term in (83) applies. Otherwise put $\bar{X} = n^{-1} \sum_{t=1}^n X_t$. Convexity gives

$$\mathbb{E}[F(\bar{X}) - F_C^*] \leq R_n/n \leq ab.$$

Let

$$q_n = \mathbb{E}[F(\bar{X}) - F_C^* \mid H_n].$$

Markov's inequality gives $\mathbb{P}(q_n \leq \gamma b) \geq 1 - a/\gamma$. On this event $r_B(H_n) \leq q_n$, so the first-hit time $\tau_b = \inf\{t \geq 1 : r_B(H_t) \leq \gamma b\}$ is at most n . Couple the policy to the first-hit block with $\tau = n \wedge \tau_b$ and $\mathbf{M} = \{\tau_b \leq n\}$. Nonnegativity of regret gives $\mathbb{E}\tau \leq n$ and $\mathbb{E}R_\tau \leq R_n$, while $\mathbb{P}(\mathbf{M}) \geq 1 - a/\gamma$. Hence (82) gives

$$b^2 n + bR_n \geq b^2 \mathbb{E}\tau + b\mathbb{E}R_\tau \geq \lambda \{I(A; Z_\tau) + d\mathbb{P}(\mathbf{M})\} \geq \lambda d(1 - a/\gamma),$$

which proves (83). Take $a = \gamma/2$ and then (84); the assumption $d\lambda/b^2 \geq 8$ absorbs the floor and gives (85). $\square$

Proof of Theorem 2.2. Fix a prior and choose a universally measurable block rule with ratio at most $1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)$ on all active scales $b \geq T^{-1}$. Theorem A.7, with $\Delta = 1$, gives (26) for every $\eta \geq T^{-1}$. Lemma C.1 replaces the resulting finite-horizon policy, for this prior, by a Borel policy with the same Bayes law. Taking the supremum over priors proves the displayed worst-prior Bayes bound. Optimizing η , with the trivial regret bound T in the complementary regime, yields (27).

Conversely, the strict inequality $\lambda < \Lambda_{\mathfrak{F}}^{\text{state}}(d, B)$ supplies an active state (Q, b) whose every first-hit experiment has ratio larger than λ . Treat Q as an initial prior and apply Proposition A.8. This gives (28) and, at the calibrated horizon, (29)–(30).

For the exponent step take $B = d^q$. Then $b \geq d^{-q}$, while the general profile bound in Proposition A.12, applied to the computation-unrestricted stochastic-BCO regret bound, gives $\Lambda_{\mathfrak{F}}^{\text{state}}(d, d^q) \leq d^{2+o(1)}$ for every subclass of bounded convex losses. Hence $\lambda \leq d^{2+o(1)}$, and the calibrated horizon is at most $d^{2q+3+o(1)}$. It lies in a polynomial horizon window; taking dimension growth rates proves (31). $\square$

A.4 Linear and quadratic calibrations

The next proposition calibrates the smallest possible universal stopped complexity. It controls arbitrary adaptive measurements, including information about joint parities or other nonlinear functions of the hidden signs.

Proposition A.9 (A linear lower bound). *Let $C = B_2^d(1)$, let $0 < b \leq 1/2$, draw $V \sim \text{Unif}\{-1, 1\}^d$, put $\theta_V = V/\sqrt{d}$, and define*

$$F_V(x) = b\{1 - \langle \theta_V, x \rangle\}. \quad (86)$$

Write Q_{lin} for this prior. Then $F_V \in [0, 2b]$, $F_V^ = 0$, and the initial Bayes risk is b . For every randomized adaptive stopped transcript Z with random length N_Z satisfying $\mathbb{E}N_Z < \infty$, and every target A obtained from (V, U) with private randomization U conditionally independent of Z given V ,*

$$I(A; Z) \leq I(V; Z) \leq \frac{b^2}{d} \mathbb{E}N_Z. \quad (87)$$

If $(Z, \mathbf{M}, x_Z)$ satisfies the posterior risk bound at level γb , then

$$\mathbb{E}N_Z \geq c_\gamma \left\{ \frac{d}{b^2} I(A; Z) + \frac{d^2}{b^2} \mathbb{P}(\mathbf{M}) \right\}. \quad (88)$$

Consequently

$$\Pi_b(Q_{\text{lin}}; A) \geq c_\gamma d. \quad (89)$$

Consequently every universal stopped-complexity upper bound must be at least of order d .

Proof. Write P_v for the law of the complete stopped transcript under $V = v$, let μ be uniform on the cube, and let $\bar{P} = 2^{-d} \sum_v P_v$. The tensorized two-point entropy inequality states that, for every positive f on the cube,

$$\text{Ent}_\mu(f) \leq \frac{1}{4} \sum_{i=1}^d \mathbb{E}_\mu[(f(V) - f(V^{(i)}))(\log f(V) - \log f(V^{(i)}))],$$

where $V^{(i)}$ flips coordinate i . Apply it pointwise to $f_z(v) = dP_v/d\bar{P}(z)$ after replacing f_z by $f_z + \varepsilon$, and let $\varepsilon \downarrow 0$; this handles densities that vanish. Integrating in z gives

$$I(V; Z) \leq \frac{1}{4} 2^{-d} \sum_{v,i} \{D_{\text{KL}}(P_v \| P_{v^{(i)}}) + D_{\text{KL}}(P_{v^{(i)}} \| P_v)\}. \quad (90)$$

For adaptive Gaussian observations, apply the finite-horizon chain rule to $N_Z \wedge m$ and let $m \rightarrow \infty$ by monotone convergence. This yields

$$D_{\text{KL}}(P_v \| P_{v^{(i)}}) = \frac{2b^2}{d} \mathbb{E}_v \sum_{t \leq N_Z} X_{t,i}^2.$$

Average this identity in (90) and use $\|X_t\|_2 \leq 1$ to obtain the second inequality in (87); the first is data processing. Since the stopped cost is at least $b^2 \mathbb{E} N_Z$, the same inequality also gives $\Xi_b(Q_{\text{lin}}) \geq d$.

On a terminal transcript $z \in \mathbf{M}$, let $m_z = \mathbb{E}[\theta_V \mid Z = z]$. The posterior risk bound gives

$$1 - \langle m_z, x_Z \rangle \leq \gamma, \quad \|m_z\|_2 \geq 1 - \gamma.$$

The law of θ_V is $1/d$ -sub-Gaussian in every unit direction. The variational formula for relative entropy therefore gives

$$D_{\text{KL}}(P_{V|Z=z} \| \mu) \geq \frac{d}{2} \|m_z\|_2^2.$$

The same calculation for any law ν on V with $r_B(\nu) \leq \gamma b$ gives $D_{\text{KL}}(\nu \| \mu) \geq d(1 - \gamma)^2/2$, and hence $\mathcal{H}_b(Q_{\text{lin}}) \geq d(1 - \gamma)^2/2$. Averaging over $z \in \mathbf{M}$ shows

$$I(V; Z) \geq \frac{d}{2} (1 - \gamma)^2 \mathbb{P}(\mathbf{M}). \quad (91)$$

Combine (87) and (91). They respectively bound $dI(A; Z)$ and $d^2 \mathbb{P}(\mathbf{M})$ by constant multiples of $b^2 \mathbb{E} N_Z$, proving (88) and then (89). $\square$

The order- d value is attained throughout the unit-ball linear class, not only on the preceding prior.

Proposition A.10 (Stopped complexity of linear losses). *Fix $0 < \beta \leq 1/2$ and consider*

$$F_\theta(x) = \beta \{1 - \langle \theta, x \rangle\}, \quad x \in B_2^d(1), \quad \theta \in S_2^{d-1}(1). \quad (92)$$

For every posterior Q on Θ , every active risk scale a with $\vartheta a < r_B(Q) \leq a$, and every admissible target coupling A , one has

$$\Pi_a(Q; A) \leq C_{\gamma, \vartheta} d. \quad (93)$$

Together with Proposition A.9, this shows that the worst stopped complexity within the class (92) is $\Theta_{\gamma, \vartheta}(d)$ up to universal constants. Thus the full-class superlinear obstruction implied by (59) must use structure absent from this fixed-amplitude Euclidean class.

Proof. Put $u = \min\{a, \beta\}$ and

$$m = \left\lceil C_0 \frac{d}{\beta u} \right\rceil,$$

where $C_0 = C_0(\gamma)$ is chosen below. Query each coordinate action e_i exactly m times. If $\bar{O}_i$ is its averaged observation, set

$$\tilde{\theta}_i = 1 - \bar{O}_i/\beta, \quad \hat{x} = \frac{\tilde{\theta}}{\|\tilde{\theta}\|_2},$$

with an arbitrary definition on the null event $\tilde{\theta} = 0$. Conditionally on every θ ,

$$\mathbb{E}_\theta \|\tilde{\theta} - \theta\|_2^2 = \frac{d}{m\beta^2}.$$

For every nonzero vector y and unit vector θ ,

$$\left\| \frac{y}{\|y\|_2} - \theta \right\|_2 \leq 2\|y - \theta\|_2,$$

because $|\|y\|_2 - 1| \leq \|y - \theta\|_2$. Therefore, uniformly in θ ,

$$\mathbb{E}_\theta F_\theta(\hat{x}) = \frac{\beta}{2} \mathbb{E}_\theta \|\hat{x} - \theta\|_2^2 \leq \frac{2d}{m\beta} \leq \frac{2u}{C_0}.$$

Choose $C_0 \geq 8/\gamma$. Under every posterior the returned action has expected regret at most $\gamma a/4$. Posterior-risk acceptance at scale a therefore gives an event $\mathbf{G}$ of probability at least $3/4$ on which $\hat{x}$ has posterior risk at most γa . Under the first-hit convention, $\mathbf{G} \subseteq \mathbf{M}$, so $\mathbb{P}(\mathbf{M}) \geq 3/4$ and actual prefix costs are no larger than the caps below.

The block has deterministic length $N = dm$, and every query has regret at most 2β , so $R_Z \leq 2\beta N$. If $a \leq \beta$, then

$$a^2 N + a \mathbb{E} R_Z \leq C_1 d^2.$$

If $a > \beta$, the origin has regret exactly β , so activity implies $\beta \geq r_B(Q) > \vartheta a$. The same display then holds with C_1 enlarged by a factor depending only on ϑ . Since the denominator of (70) is at least $d\mathbb{P}(\mathbf{M}) \geq 3d/4$, the block proves (93). The lower bound follows by taking Q to be the Rademacher prior of Proposition A.9 and $a = \beta$. $\square$

The linear calculation is not an artifact of a flat loss surface. The same exponent persists when both the minimizer and the orientation of a curved loss are unknown. This is a calculation of the stopped profile, not the first favorable-regime $d\sqrt{T}$ result for curved losses: Ito (2020) obtained optimal dimension dependence for smooth, strongly convex online BCO under an interior-minimizer condition.

Proposition A.11 (Uniformly well-conditioned quadratics have exponent one). *Fix $R \geq 1$ and, on $C = B_2^d(1)$, consider*

$$\mathfrak{Q}_R = \left\{ F_{H,y}(x) = \frac{1}{4R}(x-y)^\top H(x-y) : I \preceq H \preceq RI, \|y\|_2 \leq \frac{1}{4} \right\}. \quad (94)$$

Every member of $\mathfrak{Q}_R$ takes values in $[0, 25/64]$. For every posterior on $\mathfrak{Q}_R$, every active scale a , and every admissible private target coupling A ,

$$\Pi_a(Q; A) \leq C_{R,\gamma} d. \quad (95)$$

At a constant active scale, a shifted-hypercube subfamily satisfies, for every such target,

$$\Pi_a(Q; A) \geq c_{R,\gamma} d.$$

Consequently, for every $B \geq 64R$, both profiles of $\mathfrak{Q}_R$ are $\Theta_R(d)$. In particular, both stopped-complexity exponents equal one.

Proof. Write $F(x) = \frac{1}{2}(x-y)^\top B(x-y)$, where $\mu I \preceq B \preceq LI$ with $\mu = (2R)^{-1}$ and $L = 1/2$. Project all iterates onto $B_2^d(1/2)$, start from zero, and put $h = 1/4$. At iteration k , query $x_k \pm he_i$ once for every coordinate and use the central differences as a gradient estimate $\hat{g}_k$. All query points lie in C , and quadraticity gives

$$\mathbb{E}[\hat{g}_k \mid x_k, F] = \nabla F(x_k), \quad \mathbb{E}[\|\hat{g}_k - \nabla F(x_k)\|_2^2 \mid x_k, F] = 8d.$$

Projected stochastic gradient descent with $\eta_k = 2/\{\mu(k+k_0)\}$ and sufficiently large $k_0 = k_0(R)$ obeys the standard strongly-convex recursion

$$\mathbb{E}\|x_k - y\|_2^2 \leq C_R \min \left\{ 1, \frac{d}{k+k_0} \right\}.$$

Indeed, nonexpansiveness of projection, strong convexity, and $\|\nabla F(x)\| \leq L\|x-y\|$ give this bound by induction from

$$\mathbb{E}[\|x_{k+1} - y\|^2 \mid x_k, F] \leq (1 - 2\mu\eta_k + L^2\eta_k^2)\|x_k - y\|^2 + 8d\eta_k^2.$$

Choose k_0 so that $L^2\eta_k^2 \leq \mu\eta_k/2$. Writing $s_k = \mathbb{E}\|x_k - y\|^2$, the preceding display becomes

$$s_{k+1} \leq \left(1 - \frac{3}{k+k_0} \right) s_k + \frac{C_R d}{(k+k_0)^2}.$$

The claim follows by induction with $s_k \leq C_R d/(k+k_0)$ after enlarging C_R ; the trivial bound $s_k \leq (3/2)^2$ supplies the initial range. Taking $K = \lceil C_{R,\gamma} d/a \rceil$ makes $\mathbb{E}_F F(x_K) \leq \gamma a/16$, uniformly in F . Moreover the exact pair identity

$$\sum_{i=1}^d \{F(x_k + he_i) + F(x_k - he_i)\} = 2dF(x_k) + h^2 \text{tr}(B)$$

and the preceding decay imply

$$N_Z = 2dK \leq C_{R,\gamma} \frac{d^2}{a}, \quad \mathbb{E}_F R_Z \leq C_{R,\gamma} \frac{d^2}{a}.$$

Indeed $\mathbb{E}F(x_k) \leq C_R \min\{1, d/(k + k_0)\}$ and

$$\sum_{k=1}^K \min\left\{1, \frac{d}{k + k_0}\right\} \leq C_R d \{1 + \log(1/a)\}.$$

Thus the $2dF(x_k)$ terms contribute at most $O_R(d^2\{1 + \log(1/a)\}) = O_R(d^2/a)$. The common curvature term $h^2 \text{tr}(B) = O_R(d)$ contributes $O_R(dK) = O_R(d^2/a)$. The bounds hold under every posterior. Posterior-risk acceptance and first-hit truncation therefore give $\mathbb{P}(\mathbf{M}) \geq 15/16$. Using only the hit term in the denominator of (70) proves (95).

For the converse, fix $H = I$, draw $V \sim \text{Unif}\{-1, 1\}^d$, and set $y_V = rV/\sqrt{d}$ with $r = 1/4$. The initial Bayes action is zero and its risk is the constant $b = r^2/(4R)$. Flipping bit i changes a query mean by $-rv_i x_i/(R\sqrt{d})$. The cube-entropy comparison and stopped Gaussian chain rule used in Proposition A.9 therefore give

$$I(V; Z) \leq \frac{C_R}{d} \mathbb{E}N_Z.$$

If $m_Z = \mathbb{E}[V \mid Z]$, the terminal posterior Bayes risk is $b\{1 - \|m_Z\|^2/d\}$. Hence on $\mathbf{M}$ one has $\|m_Z\|^2 \geq (1 - \gamma)d$, and the sub-Gaussian entropy variational bound yields $I(V; Z) \geq c_\gamma d \mathbb{P}(\mathbf{M})$. Finally $I(A; Z) \leq I(V; Z)$ by data processing. Thus

$$I(A; Z) + d\mathbb{P}(\mathbf{M}) \leq \frac{C_{R,\gamma}}{d} \mathbb{E}N_Z,$$

whereas the numerator in (70) is at least $b^2 \mathbb{E}N_Z$. This proves the order- d lower bound. The explicit upper routine and this constant-scale prior give the two matching statements. $\square$

A.5 From regret bounds to stopped experiments

Proposition A.12 (A stopped experiment from a regret bound). *Suppose an algorithm has, uniformly over fixed losses and all n, δ , a high-probability regret bound*

$$R_n \leq \{D\sqrt{n} + A\} L(n, d, \delta)$$

outside an event of probability δ , where $A \geq 0$ and, for fixed constants $C_L, q < \infty$,

$$L(n, d, \delta) \leq C_L [\log\{\text{end}/\delta\}]^q. \quad (96)$$

Then, from any posterior state and scale b , it yields a first-hit experiment with

$$\mathbb{E}N_Z \leq \tilde{O}(1 + D^2/b^2 + A/b) \mathbb{P}(\mathbf{M}), \quad \mathbb{E}R_Z \leq \tilde{O}(b + D^2/b + A) \mathbb{P}(\mathbf{M}), \quad \mathbb{P}(\mathbf{M}) \geq \frac{1}{2}.$$

In particular, the computation-unrestricted bound of Fokkema et al. (2024, Theorem 2) for arbitrary bodies, and their efficient bound for symmetric bodies, have $D = Md^2 = \tilde{O}(d^{3/2})$ and $A = 0$, since in these two cases $M = \tilde{O}(d^{-1/2})$. Substitution above gives stopped complexity $\tilde{O}(D^2/d) = \tilde{O}(d^2)$ for the hit-only ratio, and hence also for the information-assisted ratio. Their efficient arbitrary-body choice $M = \tilde{O}(d^{-1/4})$ instead has $D = \tilde{O}(d^{7/4})$.

Proof. Take $\delta = \gamma b/4$ and let n be the least integer for which $DL(n, d, \delta)/\sqrt{n} + AL(n, d, \delta)/n \leq \gamma b/4$. The explicit bound (96) and a standard doubling argument give

$$n \leq C_{\gamma, C_L, q} \left(\frac{D^2}{b^2} + \frac{A}{b} + 1 \right) \left[\log \left\{ ed \left(\frac{D^2}{b^3} + \frac{A}{b^2} + \frac{1}{b} \right) \right\} \right]^{2q+2}.$$

Thus $n = \tilde{O}(1 + D^2/b^2 + A/b)$. Since regret is at most n on the failure event, $\mathbb{E}R_n \leq \gamma bn/2$. For $\bar{X} = n^{-1} \sum_{t=1}^n X_t$, convexity gives

$$\mathbb{E}[F(\bar{X}) - F_C^*] \leq \mathbb{E}R_n/n \leq \gamma b/2.$$

Apply Lemma A.3 with $a = \gamma/2$ to the terminal posterior regret of $\bar{X}$. Then $\mathbb{P}(\mathbf{M}) \geq 1/2$, so the deterministic schedule can be truncated at the first posterior state with Bayes risk at most γb . The gated terminal event implies that such a hit has occurred, while prefix length and regret can only decrease. Since $\mathbb{P}(\mathbf{M}) \geq 1/2$, the schedule cap and expected regret are absorbed by the displayed hit-probability terms. $\square$

B A multistep separation on the dimple family

The dimple construction of Bakhtiari et al. (2025, Lemma 10) makes the distinction between adaptive radii and observation-dependent direction selection exact. The following consequence is self-contained except for the known convexity of the displayed losses. It rules out a natural class of local searches; it is not a lower bound for unrestricted stochastic BCO.

Proposition B.1 (Lower bound for nonadaptive designs and fresh uniform directions). *Let d be sufficiently large, $C = B_2^d(1)$, $d^{-1} \leq \varepsilon \leq \varepsilon_0$ for a small universal constant ε_0 , $s = \sqrt{1 + 2\varepsilon}$, and $\Theta \sim \text{Unif}(S_2^{d-1}(1))$. Recall the family F_θ from (52); its losses take values in $[0, 1]$. Their prior Bayes action is $z = 0$ and*

$$\nabla F_\theta(0) = -b_0\theta. \quad (97)$$

Here b_0 is the common origin value defined in (53).

Every nonadaptive batch $X_1, \dots, X_N$, possibly randomized independently of Θ but fixed before its observations, satisfies

$$I(\Theta; O_{1:N} \mid X_{1:N}) \leq CN(\varepsilon^4 + d^{-4}). \quad (98)$$

The same bound holds sequentially for any adaptive experiment which, at time t , first draws $B_t \sim \text{Unif}(S_2^{d-1}(1))$ independently of the past and of Θ , then chooses an arbitrary history- and B_t -measurable radius $R_t \in [0, 1]$, queries $X_t = R_t B_t$, and observes $O_t = F_\Theta(X_t) + \xi_t$. For every current posterior,

$$I(\Theta; B_t, O_t \mid H_t) \leq C(\varepsilon^4 + d^{-4}). \quad (99)$$

Consequently, for this fresh-uniform-direction class, a stopped transcript Z of expected length N satisfies $I(\Theta; Z) \leq CN(\varepsilon^4 + d^{-4})$.

If an event $\mathbf{M} \in \sigma(Z)$ reports an action x_Z with posterior regret at most γb_0 , where $\gamma < 1$, then

$$I(\Theta; Z) \geq \frac{d-1}{2}(1-\gamma)^2 \mathbb{P}(\mathbf{M}). \quad (100)$$

Thus either $\Omega(d)$ information or a constant-probability target-risk hit requires

$$N = \Omega_\gamma\left(\frac{d}{\varepsilon^4 + d^{-4}}\right) = \Omega_\gamma(d/\varepsilon^4), \quad (101)$$

whereas the sample scale corresponding to complexity $O(d)$ at $b_0 \asymp \varepsilon$ is d^2/ε^2 . The restricted class loses the polynomial factor $1/(d\varepsilon^2)$ whenever $d^{-1} \lesssim \varepsilon \ll d^{-1/2}$. Its direction law, not merely its radii, must respond to the observations.

Proof. Put $q(x) = \varepsilon + \|x\|^2/2$. For $u = \langle \theta, B \rangle$,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2}(s - \sqrt{1 + r^2 - 2ru})_+^2.$$

Minimizing the square root over $r \in [0, 1]$ shows that the supremum of the right side is $(s - 1)^2/2$ when $u \leq 0$, and $(s - \sqrt{1 - u^2})^2/2$ when $u > 0$. On $|u| \leq 1/2$, its square is at most $C(\varepsilon^4 + u^8)$; the complementary spherical cap has exponentially small mass. Since

$$\mathbb{E}u^8 = \frac{105}{d(d+2)(d+4)(d+6)},$$

rotation invariance gives, uniformly in θ ,

$$\mathbb{E}_B \sup_{0 \leq r \leq 1} |F_\theta(rB) - q(rB)|^2 \leq C(\varepsilon^4 + d^{-4}). \quad (102)$$

For a nonadaptive batch, condition on its public random seed, apply the same bound after interchanging the Haar variables Θ and B by rotational invariance for each fixed design direction, and then apply the Gaussian KL bound relative to the baseline observations $q(X_t) + \xi_t$, and sum (102) over t . For the sequential class, the same estimate remains true after replacing the prior of Θ by an arbitrary posterior because the fresh B_t is uniform. Conditional Gaussian-channel capacity, with the known $q(R_t B_t)$ subtracted, proves (99); the stopped chain rule proves the transcript bound.

Convexity at the origin and (97) give

$$F_\theta(x) \geq b_0 - b_0 \langle \theta, x \rangle.$$

On an accepted transcript, with $m_Z = \mathbb{E}[\Theta \mid Z]$, posterior regret at most γb_0 therefore forces $\|m_Z\| \geq 1 - \gamma$. The uniform law σ on the sphere is $1/(d-1)$ -sub-Gaussian, so the variational formula gives

$$D_{\text{KL}}(Q \parallel \sigma) \geq \frac{d-1}{2} \|\mathbb{E}_Q \Theta\|^2.$$

Apply this to each posterior and average over $\mathbf{M}$ to obtain (100). $\square$

The next result shows that this restricted obstruction is removed when the probe directions may depend on the observations.

Theorem B.2 (Observation-adaptive experiment for the dimple family). *For the family (52), every $d^{-1} \leq \varepsilon \leq \varepsilon_0$, and all sufficiently large d , for the fixed γ in (66) there is a universal $\gamma_0 \in (0, \gamma)$ such that, for every target scale $0 < a \leq b_0$, a capped observation-adaptive experiment satisfies, uniformly over $\theta \in S_2^{d-1}(1)$,*

$$\mathcal{L}_{d,a} := 1 + \log \left\{ \frac{C_\gamma d(1 + \log(d/a))}{a} \right\},$$

and

$$\mathbb{P}_\theta \{F_\theta(\hat{x}) \leq \gamma_0 a\} \geq 1 - ca, \quad N_Z \leq \frac{C_\gamma d^2 \mathcal{L}_{d,a}^2}{a^2}, \quad \mathbb{E}_\theta R_Z \leq \frac{C_\gamma d^2 \mathcal{L}_{d,a}^2}{a}. \quad (103)$$

For every prior on Θ , applying posterior-risk acceptance to $\hat{x}$ therefore produces an event $\mathbf{G}$ of constant probability on which $\hat{x}$ has terminal posterior regret at most γa . The first-hit event $\mathbf{M}$ contains $\mathbf{G}$. For $b \leq b_0$ run the routine at $a = b$; for $b_0 < b < b_0/\gamma$ run it at $a = b_0$; and for $b \geq b_0/\gamma$, no posterior state is active because its Bayes risk is at most $b_0 \leq \gamma b$. These experiments have both stopped ratios $\tilde{O}(d)$ throughout the active range. At the critical scale $a = b_0 \asymp \varepsilon$, the adaptive cost is

$$N_Z = \tilde{O}(d^2/\varepsilon^2), \quad \mathbb{E}R_Z = \tilde{O}(d^2/\varepsilon). \quad (104)$$

The restricted lower bound in Proposition B.1 exceeds the $O(d)$ benchmark by the factor $1/(d\varepsilon^2)$ whenever that factor diverges, and the adaptive routine matches the target up to polylogarithmic factors. At the Bakhtiari–Lattimore–Szepesvári scale $\varepsilon = 8\log(d)/d$, the respective query bounds are $\tilde{O}(d^4/\log^2 d)$ and $\Omega(d^5/\log^4 d)$, a separation by a factor $\tilde{\Omega}(d)$.

At $\varepsilon = d^{-1}$, these bounds yield the following separation. Every one-query experiment is covered by the $N = 1$ nonadaptive information bound.

Proof of Theorem 2.10. For a one-query experiment, the $N = 1$ case of (98) gives the first bound below. For a fresh-uniform-direction first-hit experiment, (99) and the stopped chain rule give the second:

$$I(\Theta; Z) \leq Cd^{-4} \quad \text{or} \quad I(\Theta; Z) \leq Cd^{-4}\mathbb{E}N_Z, \quad I(\Theta; Z) \geq c_\gamma d\mathbb{P}(\mathbf{M}).$$

The last inequality gives $d\mathbb{P}(\mathbf{M}) \leq C_\gamma I(\Theta; Z)$. The sublevel-set target is a randomized function of Θ whose private coin is inaccessible, so $I(A; Z) \leq I(\Theta; Z)$. Hence

$$I(A; Z) + d\mathbb{P}(\mathbf{M}) \leq C_\gamma d^{-4}\mathbb{E}N_Z,$$

where $\mathbb{E}N_Z = 1$ in the one-query case. Since $b^2 \asymp d^{-2}$, the numerator of (70) is at least $cd^{-2}\mathbb{E}N_Z$, proving the two restricted lower bounds. The risk-reduction experiment in Theorem B.2 has $b^2\mathbb{E}N_Z + b\mathbb{E}R_Z = \tilde{O}(d^2)$ and $\mathbb{P}(\mathbf{M}) \geq c_\gamma$, which proves the unrestricted upper bound. $\square$

The proof, given in Appendix D, has three stages. A small centered orthogonal design creates correlation $\tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$. Tangent finite differences amplify this to a fixed angle. On the unit sphere the active loss has the exact form

$$F_\theta(x) = \psi(\|\theta - x\|), \quad \psi(r) = sr - r^2/2,$$

so observation-adapted boundary probes invert the tangent coordinates and halve the angular error geometrically down to $O(a)$. This last loss-sensitive stage is essential: constant correlation alone does not give loss at the target scale.

The gain comes from choosing later query directions using earlier observations; adapting only the radii does not suffice. This is a family-specific result, not a minimax lower or upper bound for the full bounded convex class.

C Technical completion: measurable implementation and duality

The results in this section close admissibility and value-level quantifiers; they do not enter the scale-amortization or first-hit arguments that drive the dimension dependence.

Lemma C.1 (Prior-wise Borel realization). *Fix a Borel prior Q and a finite horizon T . On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy that induces the same joint law of (F, H_T) under Q . The conclusion also holds when the action space is augmented by a stopping symbol.*

Proof. Represent the time- t action kernel as a universally measurable map from the history space into the standard Borel space of probability measures on the action space. Under the history marginal reached at time $t - 1$, such a map has a Borel version: apply the defining completion property to a countable generator of the target standard Borel space and use a Borel coding of that space. Replace the kernel by this version. The replacement agrees almost surely under the

reached history law and therefore leaves the joint law through time t unchanged. Induction over $t = 1, \dots, T$ gives a Borel policy with the same strategic law. A stopping decision is handled by adjoining one isolated point to the action space. In particular, every finite-horizon Bayes regret under Q is unchanged. $\square$

C.1 Bayes–minimax duality for the Gaussian model

Proof of Proposition 2.6. Let $\gamma = N(0, 1)$ and $\mathbf{H}_T = (C \times \mathbb{R})^T$. Under the reference observation law γ , let $\mathcal{S}_T$ be the set of strategic measures

$$P(dh_T) = \prod_{t=1}^T \pi_t(dx_t \mid h_{t-1}) \gamma(dy_t), \quad (105)$$

where the π_t are Borel kernels. This set is convex and weakly compact. Here and below $\mathcal{S}_T$ carries the topology of weak convergence in $\mathcal{P}(\mathbf{H}_T)$. Indeed, C is compact and every $P \in \mathcal{S}_T$ has $Y_{1:T}$ -marginal $\gamma^{\otimes T}$, so the family is tight. It is closed because, for every bounded continuous a and b ,

$$\int a(h_{t-1}, x_t) b(y_t) dP = \left(\int b d\gamma \right) \int a(h_{t-1}, x_t) dP, \quad (106)$$

and these identities pass to weak limits. Conversely, standard-Borel disintegration identifies every $P \in \mathcal{S}_T$ with a randomized nonanticipating Borel policy. Gaussian likelihoods are strictly positive, so policies identified outside a reference-null set have the same risk under every f .

For $f \in \mathfrak{F}_d(C)$ define

$$L_f(h_T) = \exp \left\{ \sum_{t=1}^T \left(y_t f(x_t) - \frac{1}{2} f(x_t)^2 \right) \right\}, \quad (107)$$

$$u(P, f) = \int L_f(h_T) \sum_{t=1}^T \{ f(x_t) - \min_C f \} dP(h_T). \quad (108)$$

The Gaussian change-of-measure formula gives $u(P, f) = \mathbb{E}_f^P R_T$. Moreover, $u(\cdot, f)$ is affine and continuous on $\mathcal{S}_T$, while $u(P, \cdot)$ is continuous in the uniform topology. The integrand is continuous and bounded by

$$T \exp \left(\sum_{t=1}^T |y_t| \right),$$

which is uniformly integrable over $\mathcal{S}_T$ because all its members have the same Gaussian $Y_{1:T}$ -marginal.

Let $\mathcal{Q}_0$ be the convex set of finitely supported priors on $\mathfrak{F}_d(C)$ and put $u(P, Q) = \int u(P, f) Q(df)$. Equip $\mathcal{Q}_0$ with the weak topology inherited from finite signed measures on the uniform-topology subspace $\mathfrak{F}_d(C)$. Since $0 \leq u(P, f) \leq T$ and $u(P, \cdot)$ is continuous, $u(P, \cdot)$ is continuous and affine on $\mathcal{Q}_0$; $u(\cdot, Q)$ is continuous and affine on $\mathcal{S}_T$. Sion's theorem (Sion, 1958), applied to the compact convex set $\mathcal{S}_T$ and the convex set $\mathcal{Q}_0$, yields

$$\min_{P \in \mathcal{S}_T} \sup_f u(P, f) = \sup_{Q \in \mathcal{Q}_0} \min_{P \in \mathcal{S}_T} u(P, Q). \quad (109)$$

The left side is minimax regret, and the right side is worst Bayes regret over finitely supported priors. Allowing all Borel priors cannot decrease the latter value and, by weak duality, cannot increase it beyond the left side. This proves (47) and the finite-support assertion. $\square$

C.2 Measurable selection of stopped experiments

We use two standard facts about relative entropy. They are recorded here because the selection argument depends on measurability of the information term and on approximation by finite prefixes.

Lemma C.2 (Relative entropy under measurable limits). *On a Polish space, $(P, Q) \mapsto D_{\text{KL}}(P\|Q)$ is lower semicontinuous, hence Borel, for the weak topology on probability laws. If $\mathcal{G}_L \uparrow \mathcal{G}$ generates the full sigma-field, then*

$$D_{\text{KL}}(P|_{\mathcal{G}_L}\|Q|_{\mathcal{G}_L}) \uparrow D_{\text{KL}}(P\|Q). \quad (110)$$

Consequently, if Z_L are increasing prefixes generating a complete transcript Z , then $I(A; Z_L) \uparrow I(A; Z)$.

Proof. The variational formula

$$D_{\text{KL}}(P\|Q) = \sup_{\phi} \left\{ \int \phi dP - \log \int e^{\phi} dQ \right\}$$

over bounded continuous ϕ proves lower semicontinuity, because each displayed functional is continuous in (P, Q) ; Borel measurability follows. Applying the same formula to bounded functions measurable with respect to $\mathcal{G}_L$, and then using martingale approximation of bounded $\mathcal{G}$ -measurable functions, proves (110). For mutual information take the two laws to be $P_{A,Z}$ and $P_A \otimes P_Z$ and restrict both to $\sigma(A, Z_L)$. $\square$

Lemma C.3 (Finite-horizon selection for Gaussian BCO). *Let C be compact and let $\mathfrak{F}_d(C)$ be a Borel subset of $\mathcal{C}(C, [0, 1])$ under the uniform topology. Fix a Borel target kernel into a standard Borel space $\mathbf{A}$ and a query cap $L < \infty$. The capped statewise value*

$$v_L(s) = \inf_{\mathcal{E}: N_Z \leq L} \rho(s, \mathcal{E}) \quad (111)$$

and the hit-only value

$$v_L^{\text{hit}}(s) = \inf_{\mathcal{E}: N_Z \leq L} \rho_{\text{hit}}(s, \mathcal{E}) \quad (112)$$

are universally measurable. For every $\epsilon > 0$, there are universally measurable state-to-experiment rules $\sigma_{L,\epsilon}$ and $\sigma_{L,\epsilon}^{\text{hit}}$ satisfying

$$\rho(s, \sigma_{L,\epsilon}(s)) \leq v_L(s) + \epsilon \quad \text{and} \quad \rho_{\text{hit}}(s, \sigma_{L,\epsilon}^{\text{hit}}(s)) \leq v_L^{\text{hit}}(s) + \epsilon \quad \text{at every active state } s. \quad (113)$$

Proof. The hidden variable for one scale is (F, A) , which takes values in a standard Borel space. Its joint posterior μ is therefore a point of the standard Borel belief space $\mathcal{P}(\mathfrak{F}_d(C) \times \mathbf{A})$. Define

$$\ell(\mu, x) = \int \{f(x) - \min_C f\} \mu(df, da), \quad r(\mu) = \min_{x \in C} \ell(\mu, x).$$

If the initial state is $s = (Q, b)$ and the target channel is $K_b(da | f)$, then

$$\mu_s(df, da) = Q(df) K_b(da | f).$$

The map $s \mapsto \mu_s$ is Borel by measurable parameter integration. The integrand is bounded and jointly continuous in (f, x) , so ℓ is Borel and continuous in x ; compactness of C makes r Borel and gives a Borel Bayes-action selector. The Gaussian observation kernel is Borel, and its strictly positive density gives the explicit Borel update

$$\Psi(\mu, x, o)(df, da) = \frac{\varphi(o - f(x)) \mu(df, da)}{\int \varphi(o - f'(x)) \mu(df', da')},$$

where φ is the standard Gaussian density. By Lemma C.2,

$$i(\mu, x) = I_\mu(A; O \mid X = x)$$

is Borel. It is finite uniformly in (μ, x) : data processing and the Gaussian capacity bound give $i(\mu, x) \leq I(F(x); F(x) + \xi) \leq \frac{1}{2} \log(5/4)$.

For fixed $\lambda > 0$, the depth- k offset problem has the belief-state recursion

$$\begin{aligned} J_k^\lambda(\mu, b) &= \inf_{x \in C} \left\{ b^2 + b\ell(\mu, x) - \lambda i(\mu, x) \right. \\ &\quad \left. + \int [-\lambda d \mathbf{1}\{r(\mu') \leq \gamma b\} + \mathbf{1}\{r(\mu') > \gamma b\} W_{k-1}^\lambda(\mu', b)] P_\mu(do \mid x) \right\}, \\ W_0^\lambda &= 0, \quad W_k^\lambda = \min\{0, J_k^\lambda\}, \quad \mu' = \Psi(\mu, x, o). \end{aligned} \quad (114)$$

The root uses J_L^λ , rather than W_L^λ , because an experiment has positive length; W_k^λ permits voluntary stopping after the first query. The posterior belief is a sufficient controlled state, so any capped history-dependent policy can be Markovized without changing its expected offset value (equivalently, the relevant belief-action occupancy measures). If τ is its stopping time, the stopped information chain rule reads

$$I(A; Z) = \mathbb{E} \sum_{t=1}^{\tau} i(\mu_{t-1}, X_t). \quad (115)$$

Together these facts show by induction that J_L^λ equals

$$\inf_{\mathcal{E}: N_Z \leq L} \{c_s(\mathcal{E}) - \lambda g_s(\mathcal{E})\}, \quad c_s = b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z, \quad g_s = I_s(A; Z) + d\mathbb{P}_s(\mathbf{M}). \quad (116)$$

Randomization causes no difficulty: its complete seed is independent of (F, A) given the state and is included in the transcript, so the offset is the average of the offsets of its deterministic realizations.

All primitive kernels and costs above are Borel, and the only negative one-step term is finite because $i(\mu, x) \leq \frac{1}{2} \log(5/4)$. For fixed L and λ the stage cost can therefore be shifted to be nonnegative before applying the standard integration theorem. The recursion preserves lower semianalyticity jointly in (μ, b, λ) , and the measurable selection theorem gives universally measurable near-minimizers at every stage, with stagewise errors allocated to sum to any prescribed total; specifically, we use the integration and partial-minimization results in Bertsekas and Shreve (1978, Propositions 7.47–7.48) and the universally measurable selector in Bertsekas and Shreve (1978, Proposition 7.50). Finally, for rational $q > 0$,

$$v_L(s) < q \iff J_L^q(\mu, b) < 0. \quad (117)$$

The right-hand side is universally measurable, so the rational strict sublevel sets in (117) first show that v_L is universally measurable. To construct the policy, enumerate $\mathbb{Q}_{>0} = \{q_1, q_2, \dots\}$. On $\{v_L < \infty\}$, choose measurably the first index $j(s)$ such that

$$v_L(s) < q_{j(s)} \leq v_L(s) + \epsilon/2,$$

and then the first $m(s)$ satisfying

$$\frac{1}{m(s)} < -\frac{1}{2} J_L^{q_{j(s)}}(\mu_s, b).$$

For each pair (j, m) , the dynamic program has a universally measurable $1/m$ -optimal offset policy. Patch these countably many policies on the sets $\{j(s) = j, m(s) = m\}$. Its induced cost and progress satisfy

$$c_s - q_{j(s)}g_s \leq J_L^{q_{j(s)}}(\mu_s, b) + \frac{1}{m(s)} < 0.$$

Thus $g_s > 0$ and $\rho(s, \sigma_{L, \epsilon}(s)) < q_{j(s)} \leq v_L(s) + \epsilon/2$. On $\{v_L = \infty\}$, any measurable one-query rule suffices and the desired inequality is vacuous. The patched stagewise rules form one universally measurable block policy on the augmented state.

For the hit-only value, use the same construction with the information increment $i(\mu, x)$ set to zero and progress $g_s^{\text{hit}} = d\mathbb{P}_s(\mathbf{M})$. Its offset recursion therefore has the same Borel primitives and lower-semianalytic structure, and the same rational sublevel and countable-patching argument gives $\sigma_{L, \epsilon}^{\text{hit}}$. This proves both inequalities in (113). $\square$

Proposition C.4 (Selection and truncation). *Let $\mathbf{S}$ be a standard Borel state space, let $\mathcal{A}_L(s)$ be the admissible first-hit experiments using at most L queries, and fix $\rho^\# \in \{\rho, \rho_{\text{hit}}\}$. Let $v_L^\#(s)$ be the corresponding capped statewise infimum and suppose that it has the measurable selectors in (113). Then the unrestricted value*

$$v^\#(s) = \inf_{\mathcal{E} \in \mathcal{A}(s)} \rho^\#(s, \mathcal{E}) \quad (118)$$

is universally measurable and has a universally measurable ϵ -optimal selector for every $\epsilon > 0$. The selected experiment may be taken to have a finite deterministic cap $L(s)$ depending on the state. These conclusions hold for either the information-assisted or the hit-only ratio. Consequently, in either case, the statewise and measurable-policy profiles coincide.

Proof. For an admissible experiment $\mathcal{E}$, stop it after at most L queries and denote the resulting first-hit experiment by $\mathcal{E}^{[L]}$. Encode each truncation as a nested transcript prefix with a stopped marker and dummy padding, so that $\sigma(Z^{[L]}) \uparrow \sigma(Z)$. Since regret is nonnegative and $\mathbb{E}N_Z < \infty$,

$$c_s(\mathcal{E}^{[L]}) \uparrow c_s(\mathcal{E}), \quad \mathbb{P}_s(\mathbf{M}^{[L]}) \uparrow \mathbb{P}_s(\mathbf{M}).$$

For the information-assisted ratio, the prefix sigma-fields increase to that of the complete almost surely finite transcript, so Lemma C.2 gives $I_s(A; Z^{[L]}) \uparrow I_s(A; Z)$. For the hit-only ratio, monotone convergence of $\mathbb{P}_s(\mathbf{M}^{[L]})$ already gives convergence of the progress. Thus, in either case, $g_s(\mathcal{E}^{[L]}) \uparrow g_s(\mathcal{E})$ and

$$\rho^\#(s, \mathcal{E}^{[L]}) \longrightarrow \rho^\#(s, \mathcal{E}), \quad (119)$$

including the zero-denominator case under the value $+\infty$ convention. It follows that $v^\# = \inf_{L \geq 1} v_L^\#$, hence $v^\#$ is universally measurable.

Choose measurably

$$L_\epsilon(s) = \min\{L : v_L^\#(s) \leq v^\#(s) + \epsilon/2\}$$

and, on the universally measurable set $\{s : L_\epsilon(s) = L\}$, use the corresponding capped selector. The patched rule stores $L_\epsilon(s)$ at block entry and is one universally measurable policy on the augmented state. It has ratio at most $v^\#(s) + \epsilon$ at every state. Taking the supremum over states and then $\epsilon \downarrow 0$ proves equality of the profiles. $\square$

Proof of Theorem 2.7. In Gaussian BCO, Lemma C.3 supplies a universally measurable near-minimizer under every finite cap. Proposition C.4 then supplies one measurable state-dependent experiment for the uncapped value, proving (48). Its hit-only part, applied to $\sigma_{L, \epsilon}^{\text{hit}}$, proves (49). $\square$

Proof of Theorem 2.8. Theorem 2.7 identifies the state and policy profiles and hence their polynomial growth exponents, separately for the information-assisted and hit-only definitions. The exponent sandwiches in Theorem 2.2 and Proposition 2.3 identify both with the Bayes exponent, while Proposition 2.6 identifies the Bayes and minimax exponents. This proves (50). $\square$

Proof of Proposition 2.9. The dense linear prior in Proposition A.9 gives the lower bound in (51). Since $\Lambda \leq \Lambda_{\text{hit}}$, this lower bound applies to both profiles. Conversely, the general BCO regret bound, inserted into Proposition A.12, constructs a constant-hit experiment and gives the upper bound for Λ_{hit} , and hence also for Λ . $\square$

D Proof of the observation-adaptive dimple experiment

Throughout this appendix fix a target $0 < a \leq b_0$ and total failure probability $\delta = c_\delta a$, where $c_\delta > 0$ is fixed below, and put $\mathcal{L} = 1 + \log\{16d(1 + \log(d/a))/\delta\}$; this agrees with $\mathcal{L}_{d,a}$ after increasing C_γ . Fix a sufficiently large universal C_{geo} , then choose $q_\star > 0$ so that

$$q_\star < s/4, \quad C_{\text{geo}}q_\star \leq s/2$$

uniformly over the stated range of ε . Next fix $\gamma_0 < \gamma$ and $c_\delta > 0$ so that $\gamma_0 + c_\delta < \gamma$, followed by the small geometric step, termination, and accuracy constants in their order of appearance and then the large repetition constants. Once chosen, these specific constants remain fixed; generic numerical constants denoted by c or C may change from line to line. We use repeatedly the standard consequence of sphere concentration that, for a Haar orthonormal basis (u_i) in a fixed k -dimensional subspace and fixed unit w ,

$$\max_i |\langle w, u_i \rangle| \leq C\sqrt{\mathcal{L}/k} \quad (120)$$

outside probability $e^{-c\mathcal{L}}$ when $\mathcal{L} \leq k$; when $\mathcal{L} > k$, we use the deterministic bound 1. All repetitions below use fresh Gaussian noise. Each stage has a deterministic query cap, the number of stages is at most $C(1 + \log(d/a))$, and confidence levels are divided across these caps so that the union of all concentration failures is at most δ .

Every use of Haar concentration is conditional on the transcript before the corresponding basis is drawn. At that point the current iterate, the hidden direction, and the activity margin are fixed, while the new basis is Haar in the stated tangent space. Thus the displayed tail bound applies conditionally at each stage. Iterated conditioning and the allocated union bound give a named good event $\mathcal{H}$ on which all activity, estimation, and contraction invariants below hold. The deterministic caps define the routine on its complement, so no assertion relies on an indefinitely repeated success event. In total, $\mathcal{H}$ intersects at most $C(1 + \log(d/a))$ conditional Haar events and at most $Cd(1 + \log(d/a))$ scalar Gaussian-mean events. The procedure is made total and Borel by terminating the capped routine and returning the origin whenever a normalized estimate is undefined, and by resolving every gate equality with the stopping branch.

Whenever $r = \|\theta - x\| < s$, expanding (52) gives the exact identity

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x \rangle - 1. \quad (121)$$

Lemma D.1 (Orthogonal initialization). *With probability at least $1 - \delta/3$, the following construction returns a unit vector v_0 satisfying*

$$\langle \theta, v_0 \rangle \geq c \min \left\{ 1, \max \left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d} \right) \right\}, \quad (122)$$

using at most $Cd^2\mathcal{L}^2/\varepsilon^2$ queries and $Cd^2\mathcal{L}^2/\varepsilon$ regret on the joint success event. If D is the noiseless centered-difference vector and ζ is its Gaussian estimation error, then

$$D = -b_0\theta + e_0, \quad \|e_0\| \leq Ch_0^2, \quad |\langle\theta, \zeta\rangle| \leq \frac{C\varepsilon}{h_0\sqrt{d\mathcal{L}}}, \quad \|\zeta\| \leq \frac{C\varepsilon}{h_0\mathcal{L}} \left(1 + \sqrt{\frac{\mathcal{L}}{d}}\right). \quad (123)$$

Proof. Draw a Haar basis $(u_i)_{i=1}^d$, set

$$h_0 = c_0 \min\{\sqrt{\varepsilon}, \varepsilon\sqrt{d/\mathcal{L}}\}, \quad n_0 = \lceil Cd\mathcal{L}^2/\varepsilon^2 \rceil, \quad (124)$$

and estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using n_0 repetitions at each of the two points. On (120),

$$h_0^2 + 2h_0 \max_i |\langle\theta, u_i\rangle| < 2\varepsilon$$

for small enough c_0 , so both points lie in the active dimple. Directly from (121),

$$D_i^0 = -\lambda_i \langle\theta, u_i\rangle, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \quad |\lambda_i - b_0| \leq Ch_0^2, \quad (125)$$

where $r_i^\pm = \|\theta \mp h_0 u_i\|$. Hence the mean difference vector is $-b_0\theta + e_0$ with $\|e_0\| \leq Ch_0^2$.

The Gaussian error vector ζ is isotropic with coordinate variance $(2n_0 h_0^2)^{-1}$. At the allocated joint failure level,

$$|\langle\theta, \zeta\rangle| \leq \frac{C\varepsilon}{h_0\sqrt{d\mathcal{L}}},$$

$$\|\zeta\| \leq \begin{cases} C\varepsilon/(h_0\mathcal{L}), & \mathcal{L} \leq d, \\ C\varepsilon/(h_0\sqrt{d\mathcal{L}}), & \mathcal{L} > d. \end{cases}$$

Let $t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}$. When $\mathcal{L} \leq d$, the definition of h_0 gives $h_0 \asymp c_0\varepsilon/t$. Hence, after first choosing c_0 small and then the repetition constant large,

$$\langle\theta, -\hat{D}^0\rangle \geq b_0 - Ch_0^2 - |\langle\theta, \zeta\rangle| \geq c\varepsilon, \quad \|\hat{D}^0\| \leq C\varepsilon/t.$$

When $\mathcal{L} > d$, one has $t = 1$; the second line of the noise bound, $\varepsilon \geq d^{-1}$, and the definition of h_0 instead give the same two inequalities with $t = 1$. On this initialization event $\hat{D}^0 \neq 0$, and we set

$$v_0 = -\hat{D}^0/\|\hat{D}^0\|, \quad \langle\theta, v_0\rangle \geq \min\left\{c_\star, c_1 \max\left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\right)\right\} =: c_{\text{init}}. \quad (126)$$

Here $c_\star > 0$ is a small universal constant. This stage uses $\tilde{O}(d^2/\varepsilon^2)$ queries. Its query points have loss $O(\varepsilon)$, so its regret is $\tilde{O}(d^2/\varepsilon)$; these are bounded by $\tilde{O}(d^2/a^2)$ and $\tilde{O}(d^2/a)$ because $a \leq b_0 \asymp \varepsilon$. $\square$

Lemma D.2 (Tangent amplification). *Conditionally on the initialization event, the capped routine in this stage reaches $\|\theta - v\| \leq q_\star$ with failure probability at most $\delta/3$, using*

$$N_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^4}, \quad R_{\text{amp}} \leq \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2}, \quad (127)$$

up to lower-order $C\mathcal{L}(1+\log(1/c_{\text{init}}))$ acceptance-test costs. At every continuing stage with correlation at least c , its probes are active, its noiseless tangent estimate g and Gaussian error ζ satisfy

$$\langle \theta, g \rangle \geq c_1 c^2, \quad \|g\| \leq Cc, \quad |\langle \theta, \zeta \rangle| \leq \frac{1}{2} c_1 c^2, \quad \|\zeta\| \leq c_{\text{amp}} c, \quad (128)$$

and the new correlation lower bound grows by a fixed factor.

Proof. Fix $q_\star$ first. Suppose a unit vector v satisfies $\langle \theta, v \rangle \geq c$, where $c_{\text{init}} \leq c \leq 1 - q_\star^2/8$. All activity and Taylor estimates below are uniform on this interval. In particular, $c^2 \geq c_2 \varepsilon$ after changing universal constants. On the unit sphere the value is the nondecreasing function

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s - q)_+^2, \quad q = \|\theta - v\|.$$

By the fixed choice of $q_\star$ above, uniformly over $0 < \varepsilon \leq \varepsilon_0$, the gap

$$\Delta_\star = \phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2)$$

is bounded below by a positive universal constant. Compare the empirical value at v with the midpoint of these two values. Gaussian concentration with $O(\mathcal{L}/\Delta_\star^2) = O(\mathcal{L})$ repetitions separates $q \leq q_\star/2$ from $q \geq q_\star$; on the overlap it may take either branch. Thus stopping implies $q \leq q_\star$, while continuing implies $q \geq q_\star/2$. On the active part the same value is the strictly increasing function

$$\psi(q) = sq - q^2/2, \quad (129)$$

for a fixed sufficiently small $q_\star > 0$.

On the latter branch, put $x = (c/2)v$ and draw a Haar tangent basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$. Write $p = \langle \theta, v \rangle \geq c$ and $r_0 = \|\theta - x\|$. Then

$$r_0^2 = 1 + \frac{c^2}{4} - cp \leq 1 - \frac{3c^2}{4}, \quad r_0 \geq 1 - \frac{c}{2}. \quad (130)$$

When $\mathcal{L} \leq d$, set $h = c_2 c$. On $\mathcal{H}$, (120) and $c \geq c_{\text{init}} \gtrsim \sqrt{\mathcal{L}/d}$ give $|q_i| := |\langle \theta, u_i \rangle| \leq Cc$. Hence

$$(r_i^\pm)^2 = r_0^2 + h^2 \mp 2hq_i \leq 1 - \frac{3c^2}{4} + h^2 + 2h|q_i| < 1 < s^2$$

after choosing c_2 sufficiently small, so every probe is active. On the capped branch $\mathcal{L} > d$, set $h = c_2 c^2$ and use $|q_i| \leq 1$ in the same display; this is again active for small c_2 . Here $c \geq \min\{c_\star, c_1\} =: c_{\text{low}} > 0$, and changing h only changes universal constants in the repetition budget. The exact central differences satisfy

$$D_i = -\lambda_i q_i, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \quad (131)$$

Writing $\lambda_0 = s/r_0 - 1$, (130), $s - 1 \asymp \varepsilon$, and $c^2 \gtrsim \varepsilon$ give $c_3 c^2 \leq \lambda_0 \leq Cc$. For completeness, apply the second-order Taylor formula to $g(t) = \sqrt{r_0^2 + t}$ at $t_\pm = h^2 \mp 2hq_i$. Because

$$\frac{t_+ + t_-}{2} = h^2, \quad \frac{t_+^2 + t_-^2}{2} = h^4 + 4h^2 q_i^2 = O(h^2),$$

and the radii are bounded below by $1/2$, the first-order odd term cancels and

$$\left| \frac{r_i^+ + r_i^-}{2} - r_0 \right| \leq Ch^2.$$

Taking reciprocals on this fixed compact interval yields

$$\lambda_0 \geq c_3 c^2, \quad \lambda_0 \leq Cc, \quad |\lambda_i - \lambda_0| \leq Ch^2. \quad (132)$$

Put

$$g = - \sum_i D_i u_i = \sum_i \lambda_i q_i u_i, \quad w = P_{v^\perp} \theta = \sum_i q_i u_i.$$

On the continuing branch, $\|\theta - v\| \geq q_*/2$, while the correlation lower bound and the choice of k below keep $\langle \theta, v \rangle$ positive. Thus $\|w\|$ is bounded below by a positive constant depending only on q_* . Equations (132) and $h \leq c_2 c$ give, for sufficiently small c_2 ,

$$\langle \theta, g \rangle = \sum_i \lambda_i q_i^2 \geq c_4 c^2, \quad \|g\|^2 = \sum_i \lambda_i^2 q_i^2 \leq Cc^2.$$

Estimate each D_i with $n_c = \lceil Cd\mathcal{L}/c^4 \rceil$ repetitions at each side. Choosing the repetition constant uniformly over the capped stages, Gaussian concentration in $v^\perp$ and the lower bound $c \geq c_{\text{init}} \gtrsim \sqrt{\mathcal{L}/d}$ (in the nontrivial $\mathcal{L} \leq d$ case) make the error ζ satisfy

$$\|\zeta\| \leq c_{\text{amp}} c, \quad |\langle \theta, \zeta \rangle| \leq (c_4/2)c^2$$

on the joint event. Thus the normalized negative estimate $u \in v^\perp$ obeys

$$\langle \theta, u \rangle \geq k_0 c \quad (133)$$

for a universal $k_0 > 0$. Choose $0 < k \leq \min\{k_0, q_*/4\}$ once and for all. Replacing v by $(v + ku)/\sqrt{1+k^2}$ raises its correlation lower bound from c to at least $\sqrt{1+k^2}c$; on the continuing branch the choice of k keeps this lower bound at most one. Iterate until it exceeds $1 - q_*^2/8$. At that point $\|\theta - v\| \leq q_*/2$, so the radial gate must stop. There are at most $C \log(2/c_{\text{init}}) = O(\log d)$ stages. The stage at correlation c uses $O(d^2\mathcal{L}/c^4)$ queries. Moreover $F_\theta(x \pm hu_i) \leq C(\varepsilon + c^2) \leq Cc^2$, so geometric summation from c_{init} gives

$$N_{\text{amp}} = \tilde{O}(d^2/\varepsilon^2), \quad R_{\text{amp}} = \tilde{O}(d^2/\varepsilon). \quad (134)$$

□

Lemma D.3 (Angular contraction). *Conditionally on the first two stage events, this capped stage returns $\hat{x}$ with $F_\theta(\hat{x}) \leq \gamma_0 a$ outside probability $\delta/3$, and has deterministic query cap $Cd^2\mathcal{L}/a^2$ and success-event regret at most $Cd^2\mathcal{L}/a$. At any intermediate scale $\|\theta - v\| \leq Q \leq q_*$, one step uses at most $Cd^2\mathcal{L}/Q^2$ queries, incurs at most $Cd^2\mathcal{L}/Q$ regret on its success event, and returns v_{new} with $\|\theta - v_{\text{new}}\| \leq Q/2$.*

Proof. Now $q = \|\theta - v\| \leq q_*$. Maintain the deterministic invariant $\|\theta - v_j\| \leq Q_j$, where $Q_j = q_* 2^{-j}$, and at one step write $v = v_j$ and $Q = Q_j$. Take $h = c_6 Q$ and draw a Haar tangent basis (u_i) of $v^\perp$. Define

$$\alpha_h = (1 + h^2)^{-1/2}, \quad t = h(1 + h^2)^{-1/2}, \quad x_i^\pm = \alpha_h v \pm t u_i.$$

All these points lie on the unit sphere and within distance CQ of θ , hence in the active dimple. Write

$$\theta = c_v v + \sum_i q_i u_i, \quad c_v = 1 - q^2/2, \quad G(p) = \psi(\sqrt{2-2p}).$$

Then exactly

$$F_\theta(x_i^\pm) = G(\alpha_h c_v \pm t q_i), \quad q_i = \frac{G^{-1}(F_\theta(x_i^+)) - G^{-1}(F_\theta(x_i^-))}{2t}. \quad (135)$$

Let

$$I_Q = \{p \in [-1, 1] : \sqrt{2-2p} \leq C_{\text{geo}} Q\}.$$

All inner products in (135) lie in I_Q . On this interval G is strictly monotone and, writing $r = \sqrt{2-2p}$, for $p < 1$,

$$G'(p) = -\frac{s-r}{r}, \quad |(G^{-1})'(G(p))| = \frac{r}{s-r} \leq CQ.$$

Indeed, $r \leq C_{\text{geo}} Q \leq C_{\text{geo}} q_\star$ and $q_\star$ was fixed so that $s-r \geq s/2$ throughout I_Q . The inverse derivative extends continuously to $p = 1$ with value zero, so the same Lipschitz bound holds at the endpoint. Project each empirical function value onto $G(I_Q)$ before applying G^{-1} . Estimating each value to accuracy $c_7 Q/\sqrt{d}$, using $\lceil Cd\mathcal{L}/Q^2 \rceil$ repetitions per site, changes each inverse inner product by at most $Cc_7 Q^2/\sqrt{d}$. Since $t \asymp Q$, every tangent coefficient changes by at most $Cc_7 Q/\sqrt{d}$. Consequently $\|\hat{w} - w\| \leq Cc_7 Q$, where $w = \sum_i q_i u_i$ and $\hat{w} = \sum_i \hat{q}_i u_i$. Clip $\hat{w}$ to norm at most $1/2$ and set

$$v_{\text{new}} = \sqrt{1 - \|\hat{w}\|^2} v + \hat{w}.$$

On the ball $\|w\| \leq 1/2$, the map $w \mapsto \sqrt{1 - \|w\|^2} v + w$ is 2-Lipschitz. With $q_\star$ fixed, choose c_7 sufficiently small. Projection onto the radius-1/2 ball is nonexpansive, so $\|v_{\text{new}} - \theta\| \leq 2Cc_7 Q \leq Q/2$.

At angular scale Q , the query count and regret are

$$N_Q = \tilde{O}(d^2/Q^2), \quad R_Q = \tilde{O}(d^2/Q), \quad (136)$$

because every probe has loss $\psi(O(Q)) = O(Q)$. Summing geometrically until $Q \leq c_{\text{term}} a$, for a sufficiently small fixed c_{term} , proves the query and regret bounds in (103). At termination, $F_\theta(\hat{x}) = \psi(\|\theta - \hat{x}\|) \leq \gamma_0 a$. $\square$

Proof of Theorem B.2. Run Lemmas D.1–D.3 in order, assigning failure probability $\delta/3$ to each. Their deterministic caps and geometric sums give

$$N_Z \leq C_\gamma d^2 \mathcal{L}_{d,a}^2 / a^2 \quad \text{and} \quad R_Z \leq C_\gamma d^2 \mathcal{L}_{d,a}^2 / a$$

on $\mathcal{H}$.

On the complement of the joint concentration event the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since its probability is at most $\delta = c_\delta a$, this adds only $C_\gamma d^2 \mathcal{L}_{d,a}^2 / a$ expected regret. The frequentist guarantee is uniform in θ , so under every prior

$$\mathbb{E}[F_\Theta(\hat{x})] \leq (\gamma_0 + c_\delta) a =: a_0 a.$$

By the choice of c_δ , $a_0 < \gamma$; apply Lemma A.3. It accepts with probability at least $1 - a_0/\gamma > 0$ and gives the required terminal posterior-risk bound. $\square$

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